Beyond leading asymptotics in the lattice nonlinear Schrödinger model

Trans-series, boundary reconstruction, and spectral determinants

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Abstract

The weak-coupling lattice nonlinear Schrödinger equation is a Love integral equation with a Lorentzian source. Its density, central rapidity distribution, and energy are point-resolvent observables rather than polynomial moments. Starting from the September 2026 analysis of Sant'Ana, we develop their beyond-all-orders description and several exact extensions. A commuting-flow representation supplies a formal endpoint seed, and first-order ordinary differential equations reduce the remaining endpoint functions to triangular recursions. A finite matrix path expansion then generates every formal sector in $\Lambda = \exp(-2\pi Q)$. We generate approximately 200 coefficients per sector and compare lateral Borel–Leroy–Padé sums with independent integral-equation solutions: the density and central-density sectors are resolved through Λ^7, while their eighth sectors remain unstable. Exact interval-variation identities determine the Fredholm determinant from the constant-source boundary value. They enforce an even-sector selection rule, yield an analytic Fisher–Hartwig normalization, and permit numerical resolution of its Λ^8 contribution. Separately, we prove a convergent boundary-to-bulk reconstruction and a regularized spectral-determinant scaling limit governed by the restricted half-Laplacian, including an explicit inverse-square spectral sum. We derive and test grand-canonical and fixed-density T^4 coefficients. Exact identities, formal resurgent constructions, and numerical evidence are distinguished throughout: a global Borel-reconstruction theorem and exponentially small corrections to individual eigenvalues are not claimed. Code, coefficients, parameter scans, and interval-certified reference values accompany the manuscript.

Keywords: lattice nonlinear Schrödinger model; Love integral equation; resurgence; Wiener–Hopf factorization; Fredholm determinants; fractional Laplacian.

Data and proof status. The supplementary archive contains executable tests and all plotted data. The endpoint spectral-contour step is a formal construction, not a uniform analytic Borel theorem. The supplied commuting-flow note by Dániel Nógrádi is cited as unpublished material.

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1 Introduction and scope

The quantum lattice nonlinear Schrödinger model, equivalently the isotropic negative-spin chain at $s = -1$, provides an integrable setting in which weak coupling leads to a singular Fredholm problem. Sant'Ana's September revision [1] resolves its inner, bulk, and edge regions by identifying the kernel with the Cauchy transition density. The central logarithm, the density constant, the Love-equation duality, and the leading restricted-half-Laplacian eigenvalue scaling are already established there. The same analysis identifies $\exp(-2\pi Q)$ as a candidate beyond-all-orders scale, while explicitly distinguishing that observation from a derivation of Borel singularities and Stokes data.

The present work starts at that distinction. Locating a zero of a Wiener–Hopf symbol does not determine which sectors occur in a given observable, their coefficients, or the prescription that reconstructs a real finite-interval solution. The difficulty is especially visible here: a point evaluation of a resolvent is not one of the polynomial moments for which a complete trans-series framework is available [2, 3, 4]. We construct the additional endpoint functions, reduce them to differential recursions, and test the resulting sectors without fitting them to the integral-equation data.

The classical Love problem [5] and its relation to the Lieb–Liniger gas [6] have a substantial analytical literature; Farina, Lang, and Martin provide a broad review [7]. High-order bulk–edge matching follows the method of Volin [8, 9], developed for weakly coupled gases by Mariño and Reis [10] and for the circular capacitor by Reichert and Ristivojevic [11]. The resurgent extensions and running-coupling conventions used below come from the Bajnok framework [12, 13, 2]. An unpublished commuting-flow note supplied with the source material [14] gives a more efficient representation of the perturbative input. We explicitly distinguish that input from the endpoint extension proposed here.

There is a complementary exact structure. Interval differentiation of a convolution resolvent produces rank-two boundary terms. These identities connect the boundary value, the determinant, and a hyperbolic boundary-to-bulk equation. This is related in spirit to the general resolvent and canonical-system constructions of Refs. [15, 16]; we do not claim that interval variation as a method is new. Its specialization here gives a particularly economical reconstruction of all lattice observables.

1.1 What is inherited and what is added

The manuscript uses version 3 of Ref. [1], dated 4 September 2026, rather than the March text. This matters for attribution. In particular, neither the density constant nor the leading $1/Q$ operator gap is an unresolved question in that revision. Likewise, the finite-temperature TBA and the gapless low-energy interpretation are already treated by Zhong et al. [17]. Our thermodynamic contribution is the reduction of higher low-temperature coefficients to the same endpoint data, together with their numerical tests.

Table 1: Scope of the continuation. "Formal" means a coefficient-generating construction whose global Borel identification remains unproved. Numerical resolution refers to the stated parameter scans, not a rigorous resummation error bound.

Result	Status and relation to earlier work
Leading density and central logarithm	Inherited from the September revision and Love-equation asymptotics.
Endpoint trans-series	Formal all-sector construction; coefficients through order 200 and sectors through Λ^{12} retained. Sectors through Λ^7 resolved numerically.
Boundary reconstruction	Exact convergent finite-interval construction from one boundary function.
Fredholm determinant	Exact boundary reduction; formal even-sector recursion; evaluated normalization; numerical resolution through Λ^8.
Low-lying spectrum	Leading fractional-Laplacian limit inherited; regularized determinant limit and spectral sum derived here. Logarithmic eigenvalue correction remains a Rayleigh-based prediction.
Low-temperature thermodynamics	Explicit T^4 endpoint formula and fixed-density conversion, checked against nonlinear thermal TBA.
Global resurgence theorem	Not established; the uniform endpoint-contour and Borel-continuation steps remain.

The term *all-sector construction* is used in a precise algorithmic sense: at a prescribed exponential order and a prescribed power of the running coupling, only finitely many differential recursions and matrix products are required.

It does not assert convergence of the infinite sector sum, a uniform error estimate, or a proof that the chosen lateral completion equals the exact solution for all positive Q.

1.2 Normalized equation and physical limits

On $[-Q, Q]$ define

$$K_Q(x, y) = K(x - y), \qquad K(x) = \frac{1}{\pi(1 + x^2)}. \tag{1.1}$$

The two basic equations are

$$(I - K_Q) f_Q = 1, \qquad (I - K_Q)\rho_Q = K. \tag{1.2}$$

The notation K_Q denotes both the integral operator and its kernel. The lattice density, central value, and dimensionless energy are

$$D(Q) = \int_{-Q}^{Q} \rho_Q(x)\mathrm{d}x = f_Q(0) - 1, \qquad R(Q) = \rho_Q(0), \qquad E(Q) = 2\pi R(Q) - 2. \tag{1.3}$$

We write R for the scalar central value and $\mathscr{R}_Q$ for the resolvent kernel, to avoid confusing them. Self-adjointness proves the density identity directly. Indeed, $K = K_Q\delta_0$, so

$$\rho_Q(x) = \mathscr{R}_Q(x, 0), \qquad \mathscr{R}_Q = K_Q(I - K_Q)^{-1} = (I - K_Q)^{-1} - I. \tag{1.4}$$

For every finite Q,

$$\|K_Q\|_{\infty\to\infty} = \frac{2}{\pi}\arctan Q < 1, \tag{1.5}$$

which proves existence, uniqueness, and positivity of the solutions.

The original coupling and Fermi rapidity obey

$$Q = \frac{q}{\kappa}, \qquad \widetilde{\rho}(x) = \kappa\rho_{\text{physical}}(\kappa x), \qquad e(\kappa, q) = -\frac{1}{\kappa}E(q/\kappa). \tag{1.6}$$

Thus $Q \to \infty$ is weak coupling at fixed q, or a high-filling limit. At fixed finite density, $D = D(Q)$ instead fixes Q and gives $q = \kappa Q$; it does not produce an additional large-Q limit. The energy sign and the factor $1/\kappa$ are kept separate from the positive function E throughout.

The inherited leading terms are

$$D(Q) = Q + \frac{\log Q}{2\pi} + \frac{1 + \log(16\pi)}{2\pi} - 1 + o(1), \tag{1.7}$$

$$R(Q) = \frac{\log(2Q) + \gamma_E}{\pi} + o(1). \tag{1.8}$$

They will fix integration constants, not be refitted in numerical comparisons.

1.3 Running-coupling dictionary

The resurgent calculation uses

$$b = \pi Q, \qquad 2b = \frac{1}{v} + \log\frac{v}{8e}, \qquad \Lambda = e^{-2b} = \frac{8e}{v}e^{-1/v}, \qquad 0 < v < 1, \tag{1.9}$$

where the small-v branch is selected. Dots denote b-derivatives:

$$\dot{F} = -\frac{2v^2}{1 - v}\partial_v F. \tag{1.10}$$

In exact Fredholm and thermodynamic formulas, primes denote Q-derivatives unless explicitly stated otherwise. The constant-source function of Ref. [2] is $2f_Q$. Its rapidity variable is $\theta = \pi x$, with endpoint b; its kernel is $[(\theta - \theta')^2 + \pi^2]^{-1}$.

The supplied O(3) note [14] uses an auxiliary variable s and flow coordinate x_{fl}. The correct dictionary is

$$s = v, \qquad x_{\text{fl}} = b + \frac{3}{2}\log 2. \tag{1.11}$$

Its physical O(3) coupling g is *not* v; $g \sim 2v$. The raw n_e, n_r coefficient file and the note's physical-coupling table provide separate checks of this distinction. Applying an additional coupling transformation and identifying its output with v would change the seed incorrectly.

2 Exact boundary control of the finite-interval problem

Define the constant-source boundary value and the lattice boundary density by

$$\chi(Q) = f_Q(Q), \qquad \eta(Q) = \rho_Q(Q). \tag{2.1}$$

The identities in this section hold for every finite positive Q. They do not require the trans-series construction of section 3. They also explain why the exact point observables contain no independent functional datum once χ is known on the whole interval of endpoint lengths.

2.1 Interval variation and the determinant

Let

$$r(Q) = \mathscr{R}_Q(Q,Q), \qquad s(Q) = \mathscr{R}_Q(Q,-Q), \qquad \mathcal{G}(Q) = \log\det(I - K_Q). \tag{2.2}$$

Domain differentiation and an integration by parts in the convolution resolvent equation give

$$\partial_Q \mathscr{R}_Q(x,y) = \mathscr{R}_Q(x,Q)\mathscr{R}_Q(Q,y) + \mathscr{R}_Q(x,-Q)\mathscr{R}_Q(-Q,y), \tag{2.3}$$

$$(\partial_x + \partial_y)\mathscr{R}_Q(x,y) = \mathscr{R}_Q(x,-Q)\mathscr{R}_Q(-Q,y) - \mathscr{R}_Q(x,Q)\mathscr{R}_Q(Q,y). \tag{2.4}$$

These are the finite-rank boundary terms familiar in Fredholm differential identities [15]. Here they immediately imply

$$r' = 2s^2, \qquad \chi' = 2s\chi, \qquad \mathcal{G}' = -2r. \tag{2.5}$$

For example, the total derivative of $\mathscr{R}_Q(Q,Q)$ includes both spatial derivatives; combining equations (2.3) and (2.4) cancels r^2 and leaves $2s^2$. Similarly,

$$\partial_Q f_Q(x) = \chi\,[\mathscr{R}_Q(x,Q) + \mathscr{R}_Q(x,-Q)], \tag{2.6}$$

$$\partial_x f_Q(x) = \chi\,[\mathscr{R}_Q(x,-Q) - \mathscr{R}_Q(x,Q)] \tag{2.7}$$

give $\chi' = 2s\chi$ at the moving endpoint.

Proposition 2.1 (Boundary formula for the determinant)**.** *For the normalized Love operator,*

$$\mathcal{G}''(Q) = -\left[\partial_Q \log\chi(Q)\right]^2, \tag{2.8}$$

and

$$\mathcal{G}(Q) = -\frac{2Q}{\pi} - \int_0^Q (Q-u)\,[\partial_u \log\chi(u)]^2 \mathrm{d}u. \tag{2.9}$$

Proof. The differential identity follows from equation (2.5). At zero interval length, $\chi(0) = 1$, $\mathcal{G}(0) = 0$, and $\mathcal{G}'(0) = -2K(0) = -2/\pi$. Twice integrating fixes both constants and gives equation (2.9). □

The parity decomposition gives a useful independent normalization test. On $[0,Q]$, let $F_\pm$ be the determinants with kernels $K(x-y) \pm K(x+y)$. Then

$$F = F_+F_-, \qquad \chi = \frac{F_-}{F_+}. \tag{2.10}$$

Indeed, $(\log F_\pm)' = -(r \pm s)$ and both determinants equal one at $Q = 0$.

The central observables obey the equally simple identities

$$D' = 2\chi\eta, \qquad R' = 2\eta^2, \tag{2.11}$$

so that

$$R' = \frac{1}{2}\left(\frac{D'}{\chi}\right)^2. \tag{2.12}$$

In the b-coordinate the last formula is

$$\dot{R} = \frac{\pi}{2}\left(\frac{\dot{D}}{\chi}\right)^2. \tag{2.13}$$

The factor of π is essential when passing between the exact and formal calculations.

2.2 A convergent boundary-to-bulk reconstruction

The complete profile satisfies a hyperbolic equation.

Theorem 2.2 (Boundary reconstruction)**.** *For $|x| < Q$,*

$$\partial_Q^2 f_Q(x) - \partial_x^2 f_Q(x) = 2\frac{\chi'(Q)}{\chi(Q)}\partial_Q f_Q(x). \tag{2.14}$$

Let $a = \chi'/\chi$, $V = a^2 - a'$, and $f_Q(x) = \chi(Q)\psi(Q,x)$. In characteristic variables $\xi = Q + x$, $\zeta = Q - x$, the unique solution is given by the convergent integral equation

$$\psi(\xi,\zeta) = 1 + \frac{1}{4}\int_0^\xi \mathrm{d}s \int_0^\zeta \mathrm{d}t\, V\Big(\frac{s+t}{2}\Big)\psi(s,t). \tag{2.15}$$

Consequently,

$$D(Q) + 1 = \chi(Q)\psi(Q,Q). \tag{2.16}$$

Proof. Write $R_\pm(x,Q) = \mathscr{R}_Q(x,\pm Q)$. Combining interval and translation differentiation yields

$$\partial_Q R_+ = -\partial_x R_+ + 2sR_-, \qquad \partial_Q R_- = \partial_x R_- + 2sR_+. \tag{2.17}$$

Differentiate equations (2.6) and (2.7), use $\chi' = 2s\chi$, and eliminate $R_\pm$. This proves equation (2.14). Substitution of $f = \chi\psi$ gives

$$\psi_{\xi\zeta} = \tfrac{1}{4}V((\xi+\zeta)/2)\psi. \tag{2.18}$$

The endpoint values $f_Q(\pm Q) = \chi(Q)$ imply $\psi(\xi,0) = \psi(0,\zeta) = 1$. Integrating twice gives equation (2.15).

On a finite rectangle where $|V| \le M$, the nth iterate of its integral operator, applied to one, has absolute value at most

$$\frac{(M\xi\zeta/4)^n}{(n!)^2}. \tag{2.19}$$

The corresponding majorant converges uniformly. The same iteration applied to the difference of two solutions proves uniqueness. Evaluation at $x = 0$ gives equation (2.16). □

This is an exact version of the boundary-to-interior philosophy in convolution and canonical-system theory [16]. In particular, it is not an asymptotic inverse problem requiring a second arbitrary integration function. For two potentials bounded by M whose uniform difference is δ, the difference of their Volterra iterates is bounded by

$$\delta\sum_{n\ge1}\frac{nM^{n-1}(\xi\zeta/4)^n}{(n!)^2}, \tag{2.20}$$

which also supplies finite-interval stability.

The central density and full lattice profile can then be recovered without another Fredholm inversion:

$$R(Q) = \frac{1}{\pi} + \frac{1}{2}\int_0^Q \left(\frac{D'(s)}{\chi(s)}\right)^2 \mathrm{d}s, \tag{2.21}$$

$$\rho_Q(x) = \eta(|x|) + \int_{|x|}^Q \frac{\eta(s)}{\chi(s)}\partial_s f_s(x)\mathrm{d}s, \qquad \eta = \frac{D'}{2\chi}. \tag{2.22}$$

For equation (2.22), vary the interval in $\mathscr{R}_Q(x,0)$ and use equation (2.6); at $Q = |x|$ the value is $\eta(|x|)$. The initial central value is $R(0) = K(0) = 1/\pi$.

Remark 2.3 (Exact information versus asymptotic seeds)**.** The exact function χ on $0 \le s \le Q$ determines the observables. A formal expansion of χ at $Q = \infty$ is not that function on the entire integration range. Inserting a divergent large-Q series into equation (2.15) down to $Q = 0$ would be invalid. The two perturbative seeds used below are therefore an algorithmic choice, not two independent exact solutions. The reconstruction supplies a route to proving their compatibility, but does not itself justify the endpoint Borel contour.

For orientation, the $Q = 0.9$ reconstruction at grid size 256 gives $D = 0.8146009425241081$, compared with $0.81460094252788799\ldots$ from the independent Fredholm solve. At $Q = 2$, size 512 gives 1.970694838377316 instead of $1.97069483838877704\ldots$. All grid sizes, including the finest-grid loss of clean Richardson scaling at $Q = 0.9$, are retained in figure 1 and the supplementary data.

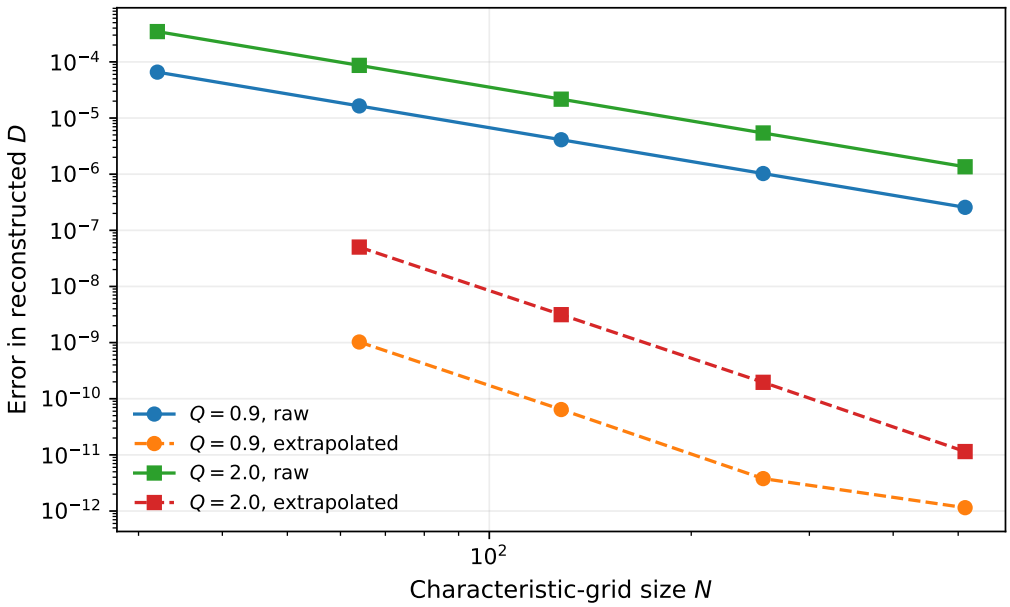


Figure 1: Independent boundary-to-bulk reconstruction. The boundary value and potential are computed from boundary resolvent columns; the central value is not an input. A second-order characteristic grid converges to the direct integral-equation value, and Richardson extrapolation accelerates the approach. The finest $Q = 0.9$ extrapolation reaches the floating-point/interpolation floor; this is a PDE-discretization test, not a trans-series-sector test.

3 An all-sector formal construction for the point observables

This section gives a finite algorithm for every coefficient of D, R, and E. Its input is the fixed-source Wiener–Hopf basis of Ref. [2] and the commuting flows of Ref. [14]. The endpoint extension is a formal construction, supported below by exact low-order checks and high-precision comparisons. We do not promote its nonuniform spectral-contour step to a theorem about Borel sums.

3.1 Why ordinary moments are not enough

Let $\chi_\alpha(\theta)$ solve the rescaled equation with source $\cosh(\alpha\theta)$, and define

$$O_{\alpha\beta}(b) = \frac{1}{2\pi}\int_{-b}^{b} \cosh(\beta\theta)\chi_\alpha(\theta)\mathrm{d}\theta = O_{\beta\alpha}(b). \tag{3.1}$$

At finite b, analytic continuation to imaginary source labels is unambiguous. The identity

$$\frac{\pi}{\theta^2+\pi^2} = \int_0^\infty e^{-\pi s}\cos(s\theta)\mathrm{d}s \tag{3.2}$$

then proves the exact transforms

$$D(Q) = 2\int_0^\infty e^{-\pi s} O_{0,is}(b)\mathrm{d}s, \tag{3.3}$$

$$R(Q) = \frac{1}{\pi} + 2\int_0^\infty\int_0^\infty e^{-\pi(s+t)} O_{is,it}(b)\mathrm{d}s\mathrm{d}t. \tag{3.4}$$

These identities relate the lattice observables to the moment framework. They do not justify termwise use of the large-b expansion near $s = t = 0$: that source limit is nonuniform. A direct imaginary-source expansion would obscure the logarithmic terms and the external residues.

The perturbative fixed-source functions satisfy [3, 2]

$$\dot{A}_{\alpha\beta} + (\alpha+\beta)A_{\alpha\beta} = a_\alpha a_\beta, \qquad \ddot{a}_\alpha + 2\alpha\dot{a}_\alpha = U a_\alpha, \qquad U = \frac{\ddot{a}_0}{a_0}. \tag{3.5}$$

The constant-source seed obeys $\dot{A}_{00} = a_0^2$. Define the endpoint transform of the perturbative basis, with the same lateral contour convention as its Wiener–Hopf factors, by

$$I_\alpha = \int_0^\infty e^{-\pi s} \left[G_+(-s) e^{isb} A_{\alpha,is} + G_-(-s) e^{-isb} A_{\alpha,-is} \right] \mathrm{d}s. \tag{3.6}$$

The perturbative endpoint contribution satisfies $I_0 = \pi D_0$. Differentiating the phases as well as the basis in equation (3.6) cancels the source label:

$$\dot{I}_\alpha + \alpha I_\alpha = a_\alpha h, \qquad h = \frac{\dot{I}_0}{a_0}. \tag{3.7}$$

Once h is known, no source integration is required. The equation at $\alpha = 0$ is an identity, however; it does not generate I_0 from A_{00}. This is the missing perturbative datum that the flow extension supplies.

3.2 Separating the zero-time seed from odd zeta values

We first summarize the supplied flow representation [14]. Set

$$t = \sqrt{1-2w}, \qquad \partial_{x_{\mathrm{fl}}} = -\frac{2w^2}{t}\partial_w, \qquad w = v + \frac{1}{2}v^3 + \frac{3}{4}v^4 + \cdots. \tag{3.8}$$

The zero-time potential and amplitude are

$$U_* = -\frac{w^2(1+w^2)}{(1-2w)^3}, \qquad a_{0,*} = \frac{1}{2}\sqrt{\frac{t}{w}}. \tag{3.9}$$

Let $L = \partial_{x_{\mathrm{fl}}}^2 - U$. The Lenard polynomials are fixed by

$$G_0 = -\frac{1}{2}, \qquad G'_{j+1} = (\partial_{x_{\mathrm{fl}}}^3 - 4U\partial_{x_{\mathrm{fl}}} - 2U')G_j, \tag{3.10}$$

with primitives zero at $w = 0$. Primes in equations (3.10), (3.18) and (3.19) mean x_{fl}-derivatives, which are identical to b-derivatives. The times are

$$\tau_{2j+1} = -\frac{4^j - 1}{(2j+1)16^j}\zeta(2j+1), \qquad j \geq 1. \tag{3.11}$$

Their action on the potential and zero-source amplitude is

$$\partial_{\tau_{2j+1}} U = -G'_{j+1}, \qquad \partial_{\tau_{2j+1}} a_0 = (2G_j\partial_{x_{\mathrm{fl}}} - G'_j)a_0. \tag{3.12}$$

The practical advantage is that one retains the flow structure rather than expanding very large polynomials of independent odd-zeta monomials at each matching step.

The endpoint seed requires the even part of the normalized spectral multiplier

$$m(k) = \sqrt{\pi}\, 2^{-k} \frac{\Gamma(1+k/2)}{\Gamma((1+k)/2)}. \tag{3.13}$$

The reflection identities for Γ give

$$\sqrt{m(k)m(-k)} = \mathcal{E}(k^2), \qquad \mathcal{E}(z) = \sqrt{\frac{\pi\sqrt{z}}{2}\cot\frac{\pi\sqrt{z}}{2}}, \quad \mathcal{E}(0) = 1. \tag{3.14}$$

The odd logarithm is

$$\frac{\log m(k) - \log m(-k)}{2} = -\sum_{j\geq 1} 4^j \tau_{2j+1} k^{2j+1}. \tag{3.15}$$

Thus the even-zeta contribution belongs to the initial endpoint function, while the odd-zeta dependence is supplied by the times. Double-counting either part changes the perturbative coefficients.

Formal construction 3.1 (Endpoint seed and flow)**.** Use the zero-time function

$$h_* = \mathcal{E}(L_*)\frac{2\sqrt{w}}{t^{3/2}}, \qquad L_* = \partial_{x_{\mathrm{fl}}}^2 - U_*. \tag{3.16}$$

The operator is expanded at zero,

$$\mathcal{E}(L) = 1 - \frac{\pi^2}{24}L - \frac{\pi^4}{640}L^2 + \cdots, \tag{3.17}$$

so every coefficient requires only finitely many differentiations. Evolve h by

$$P_j = \sum_{i=0}^{j} 4^{j-i}(2G_i\partial_{x_{\mathrm{fl}}} - G_i')L^{j-i}, \tag{3.18}$$

$$\partial_{\tau_{2j+1}} h = P_j h, \tag{3.19}$$

where the powers of L act first. Integrate

$$\dot{A}_{00} = a_0^2, \qquad \dot{D}_0 = \frac{a_0 h}{\pi}, \qquad \dot{R}_0 = \frac{h^2}{2\pi}. \tag{3.20}$$

The integration constants are fixed by the leading Love asymptotics in equation (1.7) and by

$$A_{00} = \frac{1}{16v^2} - \frac{1}{4v} - \frac{1}{16} + O(v), \qquad R_0 = \frac{-\log(\pi v) + \gamma_E}{\pi} + O(v). \tag{3.21}$$

For example, $P_1 = -4\partial^3 + 6U\partial + 3U'$. Since $La_0 = 0$, the general differential operator reduces on a_0 to the simpler flow in the supplied note. It does not reduce in that way on h. This is why the endpoint extension requires the full ordered operator.

The formal spectral calculation leading to equation (3.16) is recorded in appendix A. It combines the central contribution in the zero-time inversion with equation (3.14). The decomposition of the multiplier is exact, but the inversion and separation from endpoint-exponentially-small contributions have not been controlled uniformly. Consequently, formal construction 3.1 is not stated as a global analytic theorem. It is testable coefficient by coefficient and by its complete nonperturbative predictions.

3.3 Explicit perturbative coefficients

The first terms produced by the construction are

$$\begin{aligned} D_0(v) = {} & \frac{1}{2\pi v} - 1 + \frac{\pi^2 - 3}{12\pi}v + \frac{8\pi^2 - 9 + 27\zeta(3)}{24\pi}v^2 \\ & + \frac{9\pi^4 + 250\pi^2 - 145 + 2370\zeta(3)}{240\pi}v^3 + O(v^4), \end{aligned} \tag{3.22}$$

$$\begin{aligned} R_0(v) = {} & \frac{-\log(\pi v) + \gamma_E}{\pi} - \frac{2}{\pi}v + \frac{\pi^2 - 15}{6\pi}v^2 \\ & + \frac{40\pi^2 + 117\zeta(3) - 159}{36\pi}v^3 + O(v^4). \end{aligned} \tag{3.23}$$

These are exact symbolic results. In particular the cubic coefficient of R_0 is

$$3.328326277111553966868572666302830174249\ldots, \tag{3.24}$$

not the expression inferred from an earlier low-order numerical fit. An independent symbolic check follows by eliminating b:

$$\frac{\mathrm{d}R_0}{\mathrm{d}v} = \frac{\pi}{2}\frac{(\mathrm{d}D_0/\mathrm{d}v)^2}{\mathrm{d}A_{00}/\mathrm{d}v}. \tag{3.25}$$

The energy series follows by $E_0 = 2\pi R_0 - 2$. Re-expansion with equation (1.9) restores the required polynomials in $\log Q$; using only a constant and a single logarithm at every inverse power of Q is not a consistent high-order fitting basis.

3.4 Reduction to first-order coefficient recursions

For positive integers r, s, write

$$a_r = a_{-2r}, \quad X_r = A_{0,-2r}, \quad N_{rs} = A_{-2r,-2s}, \quad I_r = I_{-2r}. \tag{3.26}$$

They satisfy

$$\ddot{a}_r - 4r\dot{a}_r = Ua_r, \qquad a_r = 1 + O(v), \tag{3.27}$$

$$\dot{X}_r - 2rX_r = a_0 a_r, \tag{3.28}$$

$$\dot{N}_{rs} - 2(r+s)N_{rs} = a_r a_s, \tag{3.29}$$

$$\dot{I}_r - 2rI_r = a_r h. \tag{3.30}$$

For a first-order equation $\dot{F} + \lambda F = H$, with $F = v^\mu \sum f_n v^n$ and $H = v^\mu \sum h_n v^n$, the recursion is

$$f_n = \frac{1}{\lambda}\left[h_n + 2\sum_{k=0}^{n-1}(\mu + k) f_k\right]. \tag{3.31}$$

Maintaining the partial sum makes this inversion linear in the number of coefficients. The second-order equation equation (3.27) is triangular as well; the first nonzero term containing the new coefficient comes from the drift term. Homogeneous solutions growing as e^{2rb} are excluded by the prescribed formal power series. For fixed maximum sector, the expensive step is generating the scalar flow seeds; it is not the endpoint ODE inversion.

3.5 Complete internal, external, and mixed sectors

In the upper-lateral convention define

$$C_r = \frac{(r/e)^r}{(r-1)!\sqrt{2\pi r}}, \qquad S_{2r} = \frac{2\mathrm{i}(r/e)^{2r}}{(r-1)!r!} = 4\pi\mathrm{i}C_r^2. \tag{3.32}$$

The S_{2r} are the fixed-source Stokes factors of Ref. [2]. Let

$$P_{rs} = \delta_{rs} S_{2r} \Lambda^{2r}, \qquad c_r = 4\pi\mathrm{i}C_r \Lambda^r, \tag{3.33}$$

where c is the vector of external source residues. Set

$$M = (1 - PN)^{-1}P = P(1 - NP)^{-1}, \qquad J = I + Nc, \qquad Z = c + MJ. \tag{3.34}$$

All products use ordinary transposes, not complex-conjugate transposes.

Formal construction 3.2 (All-sector point observables)**.** The formal upper-lateral completion is

$$D^+ = D_0 + \frac{1}{\pi} X^T Z, \tag{3.35}$$

$$R^+ = R_0 + \frac{1}{\pi}\left[I^T c + \frac{1}{2}c^T N c + \frac{1}{2}J^T M J\right], \tag{3.36}$$

$$E^+ = 2\pi R^+ - 2. \tag{3.37}$$

The lower-lateral convention is its conjugate for real positive v. At a fixed power Λ^n, indices larger than n cannot contribute, and the inverse in equation (3.34) is a finite path expansion at that order.

Expanding $M = P + PNP + PNPNP + \cdots$ accounts for every internal path. The vector c accounts for external residues, while Nc and the matrix products generate mixed paths. In particular,

$$R_1 = 4\mathrm{i}C_1 I_1, \tag{3.38}$$

$$R_2 = 4\mathrm{i}C_2 I_2 - 8\pi C_1^2 N_{11} + \frac{S_2}{2\pi} I_1^2. \tag{3.39}$$

The direct two-source residue is only one term in the full second sector. Its constant real coefficient is e^{-2}; at the next level the corresponding real constant is $8\sqrt{2}/(3e^3)$. Both emerge automatically from the code, without insertion as input.

The external factor $4\pi\mathrm{i}C_r$ depends on the full lateral contour prescription. A half-residue replacement fails the observed ambiguity-cancellation test. The physical finite-interval equation has a unique solution, so no sector amplitude is fitted. Nevertheless, the identification of formal construction 3.2 with that unique solution after complete resummation is a separate analytic statement; it is not proved merely by choosing these constants.

3.6 Algebraic compatibility with the exact identities

The constant-source boundary amplitude and the endpoint response have the formal representations

$$\chi = a_0 + X^T M a, \qquad H = h + a^T Z. \tag{3.40}$$

If $\kappa_{rs} = 2r\delta_{rs}$, the ODEs imply

$$\dot{N} = \kappa N + N\kappa + aa^T, \quad \dot{M} = -\kappa M - M\kappa + Maa^T M, \quad \dot{Z} = -\kappa Z + MaH. \tag{3.41}$$

Together with $\dot{X} = \kappa X + a_0 a$, these identities give

$$\dot{D} = \frac{\chi H}{\pi}, \qquad \dot{R} = \frac{H^2}{2\pi}. \tag{3.42}$$

Hence equation (2.13) is obeyed sector by sector. Conversely, after generating D and χ, one can obtain R_n by the independent recursion

$$(\partial_b - 2n)R_n = \frac{\pi}{2}\sum_{k=0}^{n} U_k U_{n-k}, \qquad \frac{\dot{D}}{\chi} = \sum_{n\geq 0} \Lambda^n U_n. \tag{3.43}$$

The distributed implementation checks the matrix formula, a direct path enumeration, and equation (3.43) against one another. Their agreement is an algebraic check, distinct from comparison with TBA.

4 High-order generation and numerical validation

4.1 Coefficient generation and independent checks

The retained production calculation evaluates the commuting flows through order 210 and extracts the scalar seeds through v^{208}. The distributed point-observable files retain powers through v^{200}, with the integer and half-integer power families stored separately where both occur, at levels $0 \leq n \leq 12$. Thus there are approximately 200 coefficients per power family, not just a handful of fitted asymptotic terms.

The production implementation substitutes high-precision numerical values of the times before the high-order run, while retaining an auxiliary flow parameter. It avoids expanding separate symbolic zeta monomials at high order. It is *not* a fully symbolic 210th-order polynomial in the zeta values. The retained C++ flow run took 736.5 seconds and approximately 1.3 GB in its recorded environment; these are provenance data, not a controlled benchmark against another implementation.

There are several distinct validations.

Table 2: Independent checks and what each checks. The production coefficient generation is retained; the test suites and principal endpoint/determinant resummations were rerun when preparing this manuscript.

Comparison	Outcome and scope
A_{00} versus supplied 336-order n_e, n_r	Agreement through v^{208}; maximum recorded scaled discrepancy 2.14×10^{-161}. Tests potential, flows, and gauge, not the endpoint seed independently.
Physical-coupling table in Ref. [14]	Every displayed coefficient through g^8 reproduced, approximately 98 decimal digits numerically.
Low-order endpoint coefficients	Exact symbolic agreement with equations (3.22), (3.23) and (3.25).
Three sector implementations	Path enumeration, compact matrices, and the derivative-square recursion agree to working precision.
Independent Chebyshev TBA	Basis sizes 55 and 65 at $Q = 0.9$ differ by 1.31×10^{-62} in D and 3.29×10^{-53} in R.
Interval reference calculation	Conservative enclosures in table 3; tests the Fredholm reference independently of resummation.

The constant-source check is unusually stringent, but it must not be overinterpreted: it does not provide an independent 208-order Volin computation of the new endpoint function. The endpoint extension instead has exact low-order checks, algebraic compatibility tests, and the independent nonperturbative comparisons below.

4.2 Reference solutions and certified error bounds

For the reference integral equations we use an even-Chebyshev expansion after rescaling to $[-1, 1]$. The Lorentzian kernel acts on the modes by explicit recurrences, turning the equation into a dense linear system. This is the efficient spectral strategy used in Ristivojevic's weak-coupling calculations [18]; independent Gauss–Legendre Nyström calculations are also retained. The latter approach is standard for Fredholm determinants and their numerical control [19].

To distinguish convergence diagnostics from actual enclosures, we also compute a uniform residual certificate. If f_{app} is any explicit approximate profile and

$$r_{\mathrm{app}} = (I - K_Q) f_{\mathrm{app}} - g, \tag{4.1}$$

then equation (1.5) yields

$$\|f_{\mathrm{app}} - f\|_\infty \le \frac{\|r_{\mathrm{app}}\|_\infty}{1 - (2/\pi)\arctan Q}. \tag{4.2}$$

Interval-enclosed quadrature nodes and weights, collocation residuals, and a complex-ellipse quadrature bound give the enclosures in table 3. Details appear in appendix F. They are bounds for the reference Fredholm solutions, not for a Padé approximant.

Table 3: Certified reference values. Displayed numbers are digit prefixes. Radii apply to the full centers stored in the interval files, not to the printed truncations.

Q	Nodes	Observable	Center and enclosing radius
0.9	64	D	$0.81460094252788798855507327105240111955\ldots$ (10^{-46})
0.9	64	R	$0.53475681783401556901033074309447919970\ldots$ (10^{-46})
2	96	D	$1.97069483838877703952837171015 35\ldots$ (10^{-32})
2	96	R	$0.71509918377188479849213011102016\ldots$ (10^{-33})

4.3 Lateral Borel–Leroy–Padé summation

For each power family, factor out the lowest power v^μ and retain explicit nonpositive powers and logarithms. For the remaining series $A(v) = \sum_{k\ge 0} a_k v^k$, use

$$\mathcal{B}_\ell A(t) = \sum_{k\ge 0} \frac{a_k}{\Gamma(k+\ell)} t^k, \qquad \mathcal{S}_+ A(v) = \int_0^{\infty e^{i\phi}} e^{-t} t^{\ell-1} \mathcal{B}_\ell A(vt)\mathrm{d}t. \tag{4.3}$$

Near-diagonal Padé approximants replace the truncated Borel transform. For $\ell = 3/2$, generalized Gauss–Laguerre quadrature with weight $e^{-x}x^{1/2}$ is used after rotation $t = e^{i\phi}x$. The phase $e^{i\ell\phi}$ and the remaining factor $e^{-(e^{i\phi}-1)x}$ are included explicitly. Ordinary Laguerre quadrature without the endpoint weight creates a false numerical floor. General resurgence conventions follow Ref. [20]; the fixed-source comparisons of Ref. [2] motivate the finite-angle implementation.

Two further choices substantially improve the precision. First, resumming the primitive functions and then assembling equations (3.35) and (3.36) is more stable than separately resumming every expanded sector. Formal multiplicative compatibility does not make finite-order Padé approximants equally accurate. Second, the decaying Green solution of equation (3.30) is

$$I_r(b) = -\frac{1}{2r}\int_0^\infty e^{-y} a_r\Big(b + \frac{y}{2r}\Big)\, h\Big(b + \frac{y}{2r}\Big)\,\mathrm{d}y. \tag{4.4}$$

It avoids reconstructing by Padé the singularity generated by formal ODE inversion. Replacing the $r = 1$ endpoint block by equation (4.4) is particularly useful.

The primary $Q = 0.9$ comparison uses seed truncation $T_0 = 198$, endpoint order $T_I = 190$, fixed-source order $T_F = 130$, 512 quadrature nodes, and $\phi = 0.40$. The Green replacement uses order 190, an inner angle 0.50, 384 inner nodes, and 32 outer nodes. Working precision is 200 decimal digits, with the stated lower evaluation precision inside the nested integral recorded in the parameter file. The principal calculation was rerun from the supplied coefficients for this manuscript.

All parameter scans are retained, including failures. Small-angle 256-node quadrature generated an error floor near 10^{-13}–10^{-14} in early tests; increasing coefficient count did not cure it. The tested one-cut conformal transformations were not superior, and some near-diagonal Padé approximants were defective. No defective calculation is used as evidence for a resolved sector, and the plotted stability range is not replaced by the best single approximant.

4.4 Resolved point-observable sectors

For $F = D$ or R, define the independent remainder and prediction

$$\mathcal{R}^F_{n-1} = F_{\mathrm{TBA}} - \mathfrak{R}\mathcal{S}_+ \sum_{k=0}^{n-1} \Lambda^k F_k, \qquad \mathcal{P}^F_n = \mathfrak{R}\mathcal{S}_+(\Lambda^n F_n). \tag{4.5}$$

No TBA values are used to construct $\mathcal{P}^F_n$. The remainder includes all following sectors, so the ratio need not equal one even with exact resummation.

Table 4: Central density at $Q = 0.9$. The last line is shown to expose the present limit of the calculation, not to claim an eighth resolved sector.

n	$\mathcal{R}^R_{n-1}$	$\mathcal{P}^R_n$	Ratio
3	7.3890185×10^{-9}	7.3582717×10^{-9}	1.00417854
4	$3.0746798 \times 10^{-11}$	$3.0631079 \times 10^{-11}$	1.00377782
5	$1.1571884 \times 10^{-13}$	$1.1528718 \times 10^{-13}$	1.00374419
6	$4.3165764 \times 10^{-16}$	$4.3009538 \times 10^{-16}$	1.00363236
7	$1.5622605 \times 10^{-18}$	$1.5574887 \times 10^{-18}$	1.00306375
8	$4.7717534 \times 10^{-21}$	5.634011×10^{-21}	0.846954929 *(unresolved)*

Table 5: Total density at $Q = 0.9$, with the same independent remainder convention.

n	$\mathcal{R}^D_{n-1}$	$\mathcal{P}^D_n$	Ratio
3	$-1.3714936 \times 10^{-9}$	$-1.3683607 \times 10^{-9}$	1.0022895
4	$-3.1328639 \times 10^{-12}$	$-3.117936 \times 10^{-12}$	1.00478777
5	$-1.4927945 \times 10^{-14}$	$-1.4888193 \times 10^{-14}$	1.00267005
6	$-3.975223 \times 10^{-17}$	$-3.9594572 \times 10^{-17}$	1.00398182
7	$-1.5765847 \times 10^{-19}$	$-1.5702439 \times 10^{-19}$	1.00403812
8	$-6.340838 \times 10^{-22}$	$-4.4963235 \times 10^{-22}$	1.41022727 *(unresolved)*

The approximately 0.3%–0.4% excess in successive central-density ratios is mainly the next sector, not an error estimate for the current one. In the primary run the real errors after including $n = 8$ are 1.84×10^{-22} for D and 8.62×10^{-22} for R. These favorable values alone do not establish the eighth sector.

The stability ensemble varies $T_0 = 194, 196, 197, 198, 199, 200$ and $\phi = 0.40, 0.55$. The post-eighth-sector error divided by the magnitude of sector seven lies in

$$D: \ [-0.00143, 0.00117], \qquad R: \ [-0.00785, 0.00352]. \tag{4.6}$$

The same reasonable settings change the eighth-sector remainder/prediction ratio for R substantially, from about 0.85 to 1.88. We therefore classify sector seven as resolved at approximately the percent level or better in this ensemble, and sector eight as unresolved.

A separate, weaker-coupling test at $Q = 2$ gives $\mathcal{R}^R_4/\mathcal{P}^R_5 = 1.00001599$ for a fifth-sector contribution of approximately 1.15×10^{-28}. At this value of Q the sixth-sector signal is already below the retained resummation uncertainty. The energy ratios are identical to those of R, because $E_n = 2\pi R_n$ for $n \geq 1$; they are not a third independent validation.

The observation that principal sums agree to many digits should not be conflated with a proof of an infinite resummed trans-series. The certified reference error is negligible at the resolved sector scales; the remaining uncertainty is the analytic-continuation and finite-order Padé error on the trans-series side.

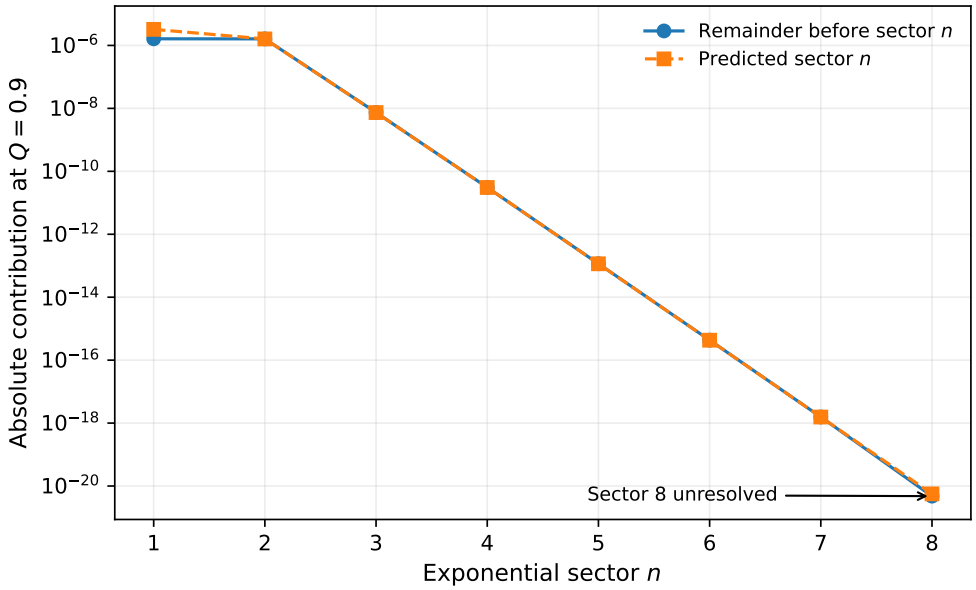


Figure 2: Central-density hierarchy at $Q = 0.9$. Absolute values are used only in the plot; the signed values are in table 4. The first two lateral sectors undergo strong mutual cancellations and should not be interpreted as independent positive contributions. Successive remainders track the independently generated sectors through $n = 7$. The eighth is not stable.

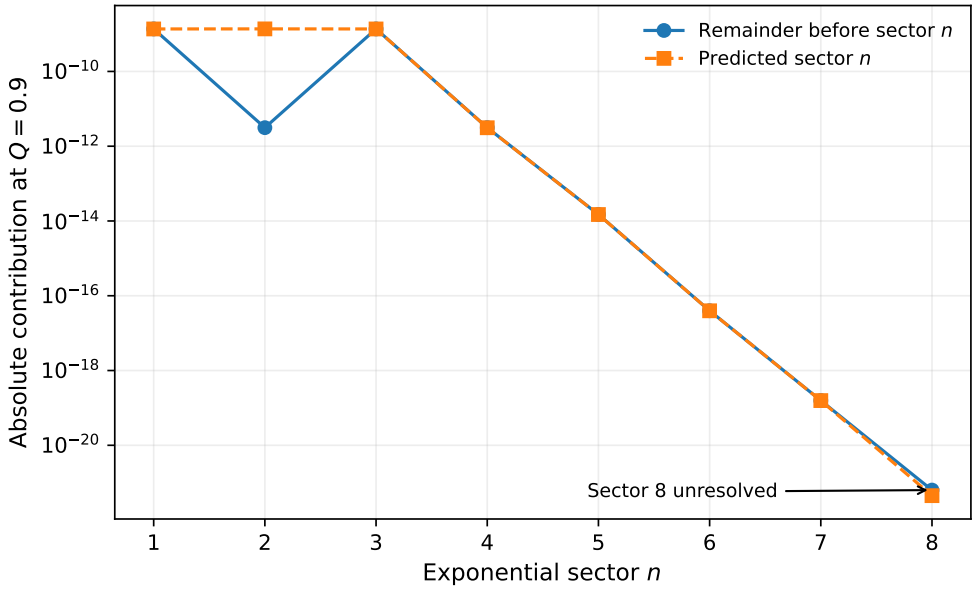


Figure 3: Density hierarchy, using the same parameters and definitions as figure 2. The seventh contribution is approximately -1.57×10^{-19} and is resolved against the reference solution. Agreement of the eighth contribution in scale alone is insufficient for resolution.

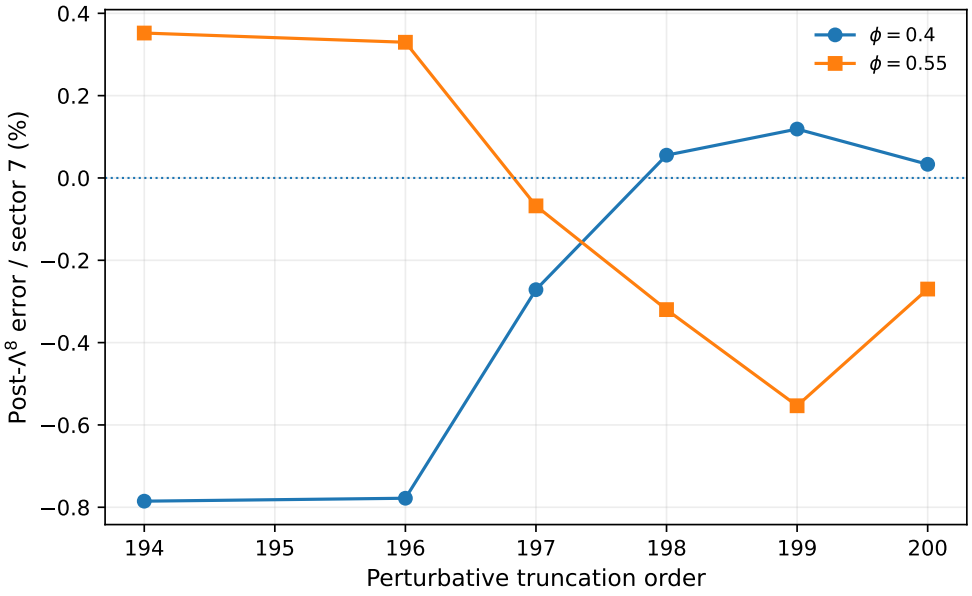


Figure 4: Central-density stability diagnostic. The remaining real error after eight sectors is normalized by the magnitude of sector seven. Both contour-angle families and all retained truncations are shown. The spread is small compared with sector seven but not with sector eight. These are empirical diagnostics, not interval bounds on Borel summation.

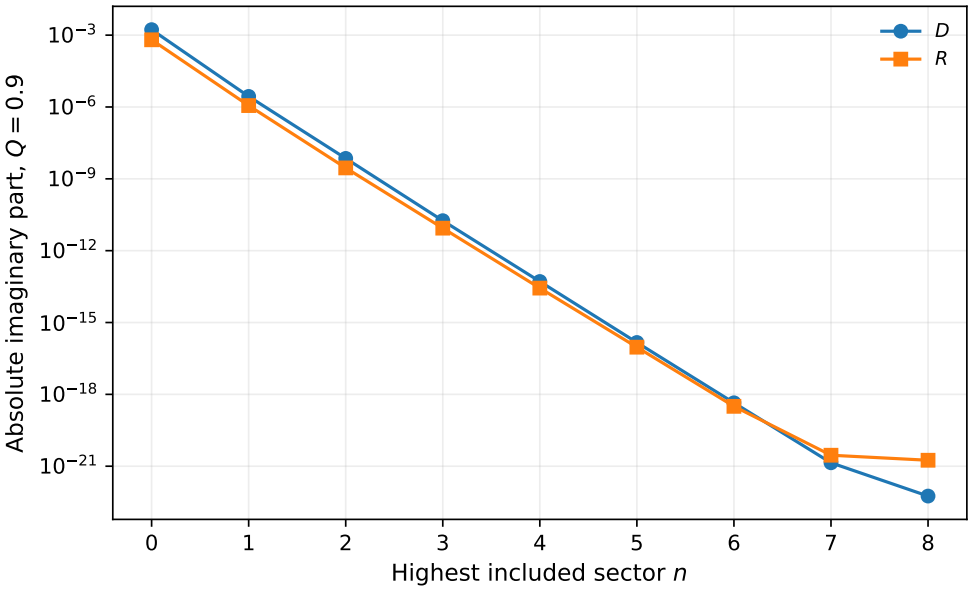


Figure 5: Cumulative imaginary parts of the upper-lateral approximations. The cancellation from 10^{-3}–10^{-4} to approximately 10^{-21}–10^{-22} is sensitive to the external-residue normalization. It is an independent consistency check, not a substitute for the signed real-remainder comparison.

5 The Fredholm determinant and its sector selection rule

The determinant is simpler than the point observables in one important respect: proposition 2.1 needs only the constant-source boundary value χ, not the new endpoint seed h. Its formal nonperturbative content is therefore inherited from the fixed-source path construction.

5.1 A complete even-sector recursion

The boundary amplitude in equation (3.40) contains only even powers of Λ, since $P_{rr} \propto \Lambda^{2r}$ and no external residue c occurs. In the b-coordinate, equation (2.8) has the same form:

$$\ddot{\mathcal{G}} = -(\partial_b \log \chi)^2. \tag{5.1}$$

Expand

$$p = \partial_b \log \chi = \sum_{n\geq 0} \Lambda^n p_n, \qquad \mathcal{G} = \sum_{n\geq 0} \Lambda^n \mathcal{G}_n. \tag{5.2}$$

For every $n > 0$,

$$(\partial_b - 2n)^2 \mathcal{G}_n = -\sum_{k=0}^{n} p_k p_{n-k}. \tag{5.3}$$

Applying equation (3.31) twice determines the sector; growing homogeneous terms are excluded. It follows that

$$\mathcal{G}_{2r+1} = 0. \tag{5.4}$$

Thus the first allowed formal determinant sector is $\Lambda^2 = e^{-4\pi Q}$, even though the elementary external-source scale for the lattice point observables is $\Lambda = e^{-2\pi Q}$. This refines, rather than contradicts, an ansatz that allows arbitrary multiples of the candidate scale: its odd determinant coefficients vanish in this convention.

Exact symbolic arithmetic gives

$$\mathcal{G}_2 = -\frac{\mathrm{i}}{e^2}\left(\frac{v}{2} + \frac{5v^2}{8} + \frac{61v^3}{64} + \cdots\right), \tag{5.5}$$

$$\mathcal{G}_4 = \frac{1}{e^4}\left[\frac{1}{4} + \left(\frac{1}{8} - \mathrm{i}\right)v + \left(\frac{7}{32} - \frac{13\mathrm{i}}{8}\right)v^2 + \cdots\right], \tag{5.6}$$

$$\mathcal{G}_6 = \frac{1}{e^6}\left[\frac{16}{9} + \left(\frac{2}{3} - \frac{13\mathrm{i}}{4}\right)v + \left(\frac{9}{8} - \frac{91\mathrm{i}}{16}\right)v^2 + \cdots\right]. \tag{5.7}$$

The purely imaginary formal coefficients in equation (5.5) do not permit that sector to be omitted. The real part of a lateral sum is not the lateral sum of coefficientwise real parts. Exponentiating $\mathcal{G}$ gives the corresponding determinant sectors.

5.2 Perturbative normalization and the Fisher–Hartwig constant

The symbol is $\sigma(p) = 1 - e^{-|p|}$. Its root at zero produces the logarithmic term in the Wiener–Hopf/Fisher–Hartwig asymptotics [21, 22]. In the interval-length convention $2Q$, write

$$\mathcal{G}_0(Q) = -\frac{\pi Q}{3} + \frac{1}{4}\log Q + \beta_{\mathrm{FH}} + \cdots. \tag{5.8}$$

The curvature equation does not fix the asymptotic slope and constant by local large-Q integration. The extensive term follows from the symbol integral; the constant requires the Fisher–Hartwig normalization or the exact finite-interval formula equation (2.9).

Evaluating the single-root normalization gives

$$\boxed{\beta_{\mathrm{FH}} = -\frac{1}{6} + \frac{7}{12}\log 2 + \frac{3}{4}\log \pi + 6\zeta'(-1).} \tag{5.9}$$

Numerically,

$$\beta_{\mathrm{FH}} = 0.10368974084431265248410462290871317519529457\ldots. \tag{5.10}$$

The calculation uses the Fourier transform

$$g(t) = \frac{1}{2\pi t^2} - \frac{\coth(\pi t)}{2t} \tag{5.11}$$

and the regularized integral

$$C_E = \lim_{L\to\infty}\left[\int_0^L t g(t)^2 \mathrm{d}t - \frac{1}{4}\log L\right] = \frac{\gamma_E}{4} - \frac{1}{6} + \frac{1}{4}\log(2\pi) + 3\zeta'(-1). \tag{5.12}$$

The Barnes-G factor and the length conversion yield equation (5.9); details are in appendix C. This is an analytic evaluation under the standard single-root normalization, not a fit. A self-contained verification of every limiting smooth-factor hypothesis of the general Fisher–Hartwig theorem is not supplied here. The independent determinant comparison is an especially sensitive check of the normalization.

The perturbative expansion is

$$\mathcal{G}_0 = -\frac{b}{3} - \frac{1}{4}\log v + \beta_{\mathrm{FH}} - \frac{1}{4}\log(2\pi) - \frac{1}{8}v - \frac{11}{48}v^2 + \frac{27\zeta(3) - 109}{288}v^3 + \cdots . \tag{5.13}$$

In terms of $L_Q = \log(16\pi Q)$,

$$\mathcal{G}_0 = -\frac{\pi Q}{3} + \frac{1}{4}\log Q + \beta_{\mathrm{FH}} + \frac{2L_Q + 1}{16\pi Q} + \frac{-6L_Q^2 + 6L_Q + 1}{192\pi^2 Q^2} + \cdots . \tag{5.14}$$

The September source gives the leading determinant form with an unspecified constant. Equations (5.9) and (5.3) are the normalization and sector refinements needed to make the comparison quantitative.

5.3 Independent determinant comparison

We retain 180 perturbative orders per determinant sector through Λ^8. The exact comparison uses parity-resolved arbitrary-precision Nyström determinants, not integration of the trans-series for χ. At $Q = 0.9$,

$$\mathcal{G}_{\mathrm{num}} = -0.70869442017436008947006421806477777174 17\ldots . \tag{5.15}$$

Quadrature sizes 65 and 80 agree to the 65-digit output precision, and the ratio F_-/F_+ agrees independently with the boundary value. The principal Borel calculation uses 154 terms, angle 0.45, 512 nodes, and 170-digit arithmetic; it was rerun in preparing this manuscript.

Table 6: Determinant logarithm at $Q = 0.9$. The remainder is taken after all preceding even sectors, including Λ^2, have been summed.

Sector	Exact remainder	Predicted real contribution	Ratio
Λ^4	$7.24113764791 \times 10^{-13}$	$7.24105335134 \times 10^{-13}$	1.00001164
Λ^6	$8.42965686761 \times 10^{-18}$	$8.42956661378 \times 10^{-18}$	1.00001071
Λ^8	$9.02538365767 \times 10^{-23}$	$9.02341388945 \times 10^{-23}$	1.00021830

After including Λ^8, the primary real error is 1.97×10^{-26} and the cumulative imaginary part is -4.04×10^{-26}. The additional settings $(150, 0.55)$ and $(158, 0.40)$, with the same 512 nodes, put the last remainder/prediction ratio in $[1.0001683, 1.0003237]$. This resolves the eighth determinant sector at a few parts in 10^4 in the tested ensemble. It does not resolve the eighth sectors of D or R, which are numerically different problems.

The coefficient of Λ^2 has a real lateral contribution comparable to the subsequent Λ^4 contribution. Accordingly, its remainder/prediction ratio alone is close to one half rather than one. This is another reason to retain complex lateral sectors and test the complete sequence of cancellations. The complete table, including the perturbative and second sectors, is supplied rather than hidden.

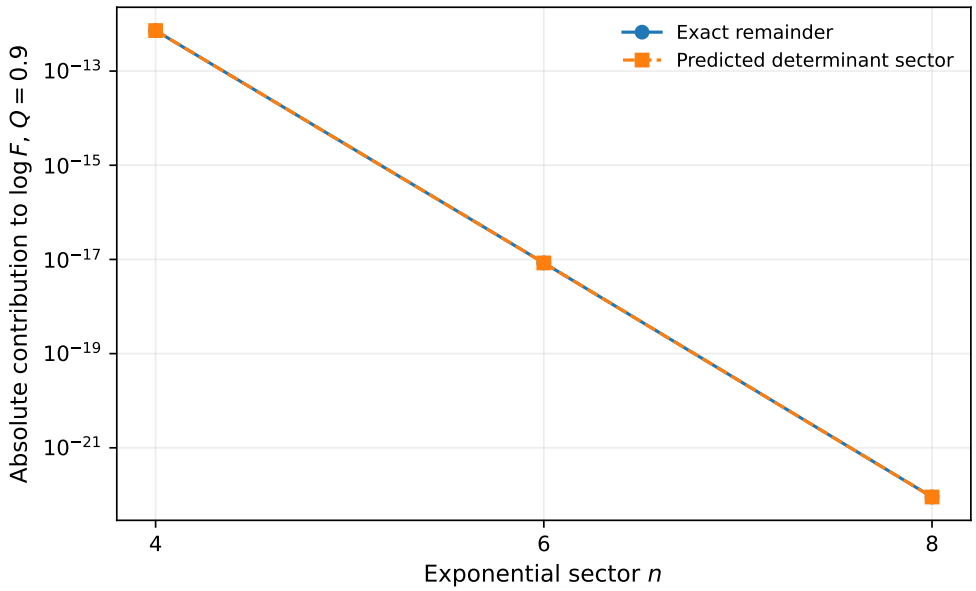


Figure 6: Independent determinant test. The $\Lambda^8 = e^{-16\pi Q}$ contribution is visible after subtraction of the preceding sectors. Agreement is not inferred from the position of a symbol zero: both the amplitude and fluctuation series are predicted by equation (5.3).

6 A uniform spectral-determinant limit and spectral refinements

Let $1 > \mu_0(Q) \geq \mu_1(Q) \geq \cdots > 0$ be the eigenvalues of K_Q. They are operator eigenvalues, not physical excitation energies. Upon scaling $x = Qy$ and setting $h = 1/Q$, the operator becomes

$$T_h = P_{(-1,1)} e^{-h|\partial|} P_{(-1,1)}. \tag{6.1}$$

This is a compression of the full-line Cauchy semigroup. At nonzero h it is not the continuously killed interval semigroup. The limiting generator is the *restricted* half-Laplacian A: functions are extended by zero outside $(-1, 1)$ before the Fourier multiplier $|p|$ is applied. The spectral square root of the ordinary Dirichlet Laplacian is a different operator.

Sant'Ana's September revision already identifies the fixed-mode limit

$$Q[1 - \mu_{j-1}(Q)] \longrightarrow \lambda_j, \tag{6.2}$$

where λ_j are the eigenvalues of A. The fractional-operator spectral input is established in Ref. [23]. The new objective is a simultaneous analytic object containing all these limits, suitable for a later uniform eigenvalue trans-series.

6.1 Regularized deformed determinant

Define

$$F(z, Q) = \det(I - zK_Q), \qquad B_Q = \frac{1}{Q} K_Q (I - K_Q)^{-1}. \tag{6.3}$$

Factoring the two operators gives the exact identity

$$\mathfrak{D}_Q(u) := \frac{F(1 + u/Q, Q)}{F(1, Q)} e^{u \operatorname{Tr} B_Q} = \det{}_2(I - uB_Q), \tag{6.4}$$

where

$$\det{}_2(I - uB) = \prod_{j \geq 1} (1 - u\beta_j) e^{u\beta_j}. \tag{6.5}$$

The exponential subtraction cannot be omitted: the limiting inverse is Hilbert–Schmidt but not trace class.

Theorem 6.1 (Uniform spectral-determinant limit)**.** *As $Q \to \infty$,*

$$\mathfrak{D}_Q(u) \longrightarrow \det{}_2(I - uA^{-1}) \tag{6.6}$$

locally uniformly for complex u. In particular, its fixed low zeros

$$u_j(Q) = Q[\mu_{j-1}(Q)^{-1} - 1] \tag{6.7}$$

converge to the corresponding eigenvalues λ_j.

Proof. For zero-extended f, the quadratic forms

$$q_h(f) = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{1 - e^{-h|p|}}{h} |\widehat{f}(p)|^2 \mathrm{d}p \tag{6.8}$$

increase to the form of A as $h \downarrow 0$. The min–max upper bound follows by testing on fixed eigenfunctions of A. Bounded q_h controls the Fourier tail: for fixed M and sufficiently small h, its mass above M is at most $2C/M$. Together with the fixed spatial support, this yields compactness of bounded-energy eigenvectors. The lower variational bound then proves equation (6.2). Consequently,

$$\beta_j(Q) = \frac{\mu_{j-1}(Q)}{Q[1 - \mu_{j-1}(Q)]} \longrightarrow \lambda_j^{-1}. \tag{6.9}$$

It remains to control the product uniformly in j. Positivity of the Cauchy kernel permits dropping the restrictions on intermediate integration points in a trace cycle, giving

$$\operatorname{Tr} T_h^m \le \frac{2}{\pi m h}, \qquad m = 1, 2, \dots. \tag{6.10}$$

Since $j\mu_{j-1}^m \le \operatorname{Tr} T_h^m$, choosing $m = \lceil 1/(-\log \mu_{j-1}) \rceil$ when $-\log \mu_{j-1} \le 1$, and $m = 1$ otherwise, yields

$$0 \le \beta_j(Q) \le \frac{2e^2}{\pi j}. \tag{6.11}$$

The squared majorant is summable. The tails of the regularized products therefore converge uniformly on compact u-sets, while every finite head converges termwise. This proves equation (6.6). Zero convergence follows either from the explicit u_j formula or from local uniform convergence with multiplicities. □

The theorem is a scaling result, not a finite-Q trans-series for individual eigenvalues. It identifies the correct spectral window $z = 1 + u/Q$. A trans-series evaluated only at $z = 1$ cannot determine the displaced zeros in that window.

6.2 An exact inverse-square spectral sum

The published Green kernel of the restricted half-Laplacian is [23]

$$G_A(x, y) = \frac{1}{\pi} \log \frac{1 - xy + \sqrt{1 - x^2}\sqrt{1 - y^2}}{|x - y|}. \tag{6.12}$$

Squaring and integrating gives an explicit trace.

Proposition 6.2.

$$\operatorname{Tr} A^{-2} = \sum_{j \ge 1} \lambda_j^{-2} = \frac{2 + 7\zeta(3)}{\pi^2} = 1.055199164919687124734801779611789\dots. \tag{6.13}$$

Proof. Set $x = \tanh t$, $y = \tanh s$. The logarithm in equation (6.12) becomes $\log \coth(|t - s|/2)$. The convolution identity

$$\int_{\mathbb{R}} \operatorname{sech}^2 t \operatorname{sech}^2(t - u) \mathrm{d}t = 4(u \coth u - 1) \operatorname{csch}^2 u \tag{6.14}$$

reduces the trace to a single integral. With $z = \tanh(u/2)$ this is

$$\operatorname{Tr} A^{-2} = \frac{4}{\pi^2} \int_0^1 [(z^{-3} - z) \operatorname{atanh} z - z^{-2} + 1] \log^2 z \mathrm{d}z. \tag{6.15}$$

Expand $\operatorname{atanh} z$ after the displayed cancellation of the singular term. Integrating each monomial and summing the resulting partial fractions gives $1/2 + 7\zeta(3)/4$ for the integral. The explicit summation is given in appendix D. □

An independent quadrature agrees to approximately 63 decimal digits. We use the sum as an analytic normalization and numerical check; no claim of first publication of this identity is needed. It fixes

$$\log\det{}_2(I - uA^{-1}) = -\frac{2 + 7\zeta(3)}{2\pi^2}u^2 + O(u^3). \tag{6.16}$$

For efficient spectral computation, an energy-orthonormal basis is

$$e_n(x) = \sqrt{\frac{2}{\pi n}}\sin(n\arccos x), \qquad n \geq 1. \tag{6.17}$$

The compact embedding matrix, isospectral to A^{-1}, has entries

$$C_{mn} = \frac{2}{\pi\sqrt{mn}}\left[\frac{1}{1-(m-n)^2} - \frac{1}{1-(m+n)^2}\right] \tag{6.18}$$

for equal parities and zero for opposite parities. This follows from $A[\sqrt{1-x^2}U_{n-1}(x)] = nU_{n-1}(x)$ and direct trigonometric integration. It avoids singular-kernel quadrature for the limiting operator.

Table 7: Finite-Q spectral scaling. The listed finite values use the higher retained panel quadrature order. Limiting fractional eigenvalues are numerical Ritz estimates, whereas the limiting trace is the exact expression in equation (6.13).

Q	First zero $u_1(Q)$	$\operatorname{Tr} B_Q^2$
16	1.141241417292	0.990942224358
64	1.149941515833	1.032868751319
256	1.154835298067	1.048009457445
∞	1.157773883698...	1.055199164920...

6.3 The leading gap and its logarithmic refinement

For completeness, the elementary bounds

$$1 - \frac{2}{\pi}\arctan Q \leq 1 - \mu_0(Q) \leq \frac{3\pi}{8Q} \tag{6.19}$$

follow from the maximal row sum and the trial function $\sqrt{1-y^2}$, respectively. Together with equation (6.2), they confirm the $1/Q$ scale already identified in the September revision. The first fractional eigenvalue is approximately 1.15777388369759; the unnormalized operator gap of Ref. [1] is $2\pi(1-\mu_0)$.

There is a further logarithm at the next algebraic order. For an L^2-normalized A-eigenfunction, let

$$\phi_j(x) \sim c_\pm\sqrt{\operatorname{dist}(x, \partial(-1,1))} \tag{6.20}$$

at the two endpoints. The fractional Pohozaev identity [24] gives

$$c_+^2 + c_-^2 = \frac{4\lambda_j}{\pi}. \tag{6.21}$$

Its nonoscillatory Fourier tail is $\pi(c_+^2 + c_-^2)/(4|p|^3)$. Inserting this into equation (6.8) gives, for the *fixed limiting eigenfunction*,

$$q_h(\phi_j) - q(\phi_j) = -\frac{\lambda_j}{2\pi}h\log(1/h) + O(h). \tag{6.22}$$

The oscillatory cross-endpoint term does not produce the logarithm.

Conjecture 6.3 (First actual-eigenvalue correction)**.** *For each fixed mode,*

$$1 - \mu_{j-1}(Q) = \frac{\lambda_j}{Q} - \frac{\lambda_j}{2\pi}\frac{\log Q}{Q^2} - \frac{C_j}{Q^2} + o(Q^{-2}). \tag{6.23}$$

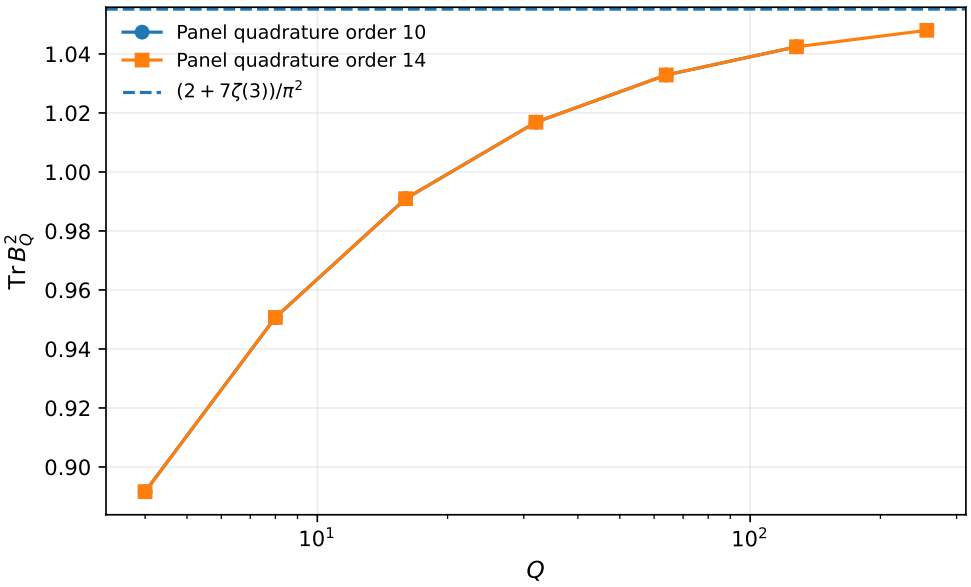


Figure 7: The quadratic coefficient of the regularized spectral determinant approaches the exact inverse-square spectral sum. Two panel quadrature orders are retained. This is a test of the spectral scaling theorem, not an exponentially small eigenvalue correction.

The logarithmic coefficient in equation (6.22) is derived, but promotion to equation (6.23) requires a perturbed-eigenfunction estimate that is not supplied. It would be incorrect to identify the Rayleigh calculation with a proof of the actual eigenvalue expansion.

We test the predicted, rather than fitted, logarithmic coefficient for four modes using a Toeplitz/Galerkin eigensolver, two cell resolutions, and Richardson extrapolation. The data extend through $Q = 4096$ for four modes and through 8192 for the first gap. After subtracting the predicted logarithm, the first constant estimate increases from 1.04913 at $Q = 128$ to 1.06915 at $Q = 4096$. The other modes stabilize similarly; the constants themselves are not assigned closed forms.

Individual exponentially small eigenvalue corrections remain uncomputed. Before extracting them, the algebraic and logarithmic background must be controlled uniformly near the zeros of equation (6.4). The determinant trans-series at $z = 1$ alone cannot provide that control.

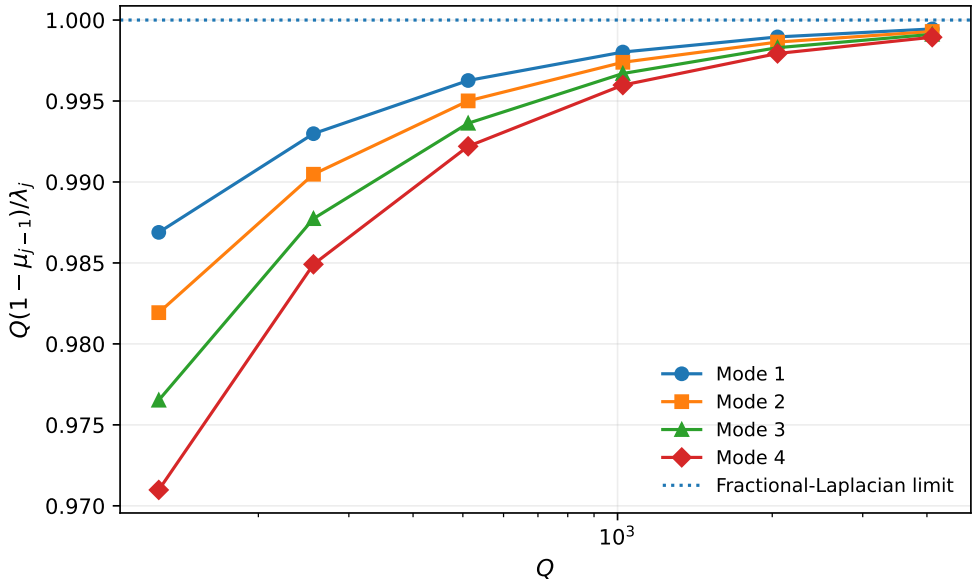


Figure 8: Four low modes approach the known restricted-half-Laplacian limit. The normalization is by the independently computed λ_j. The figure checks the inherited leading scale and supplies the data used for the refinement in figure 9.

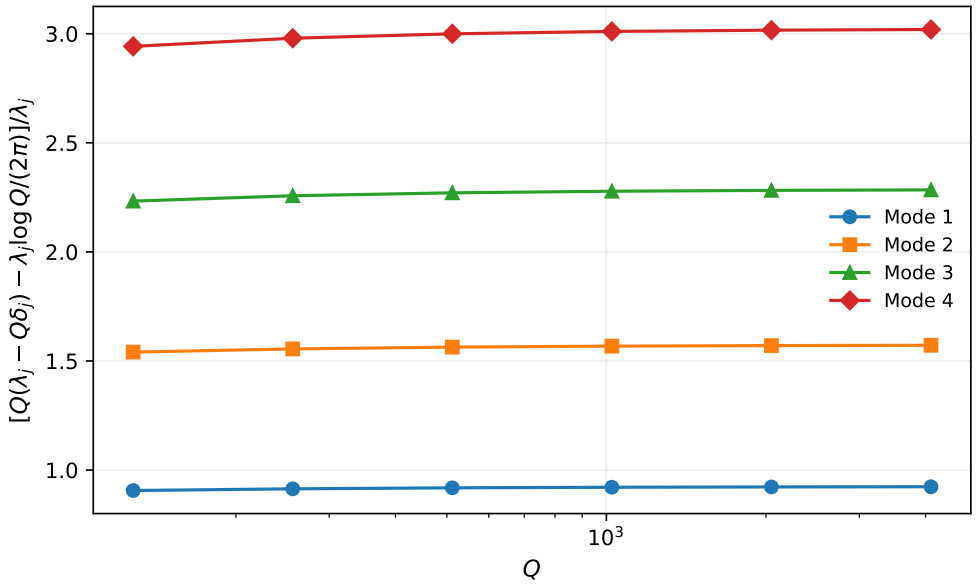


Figure 9: After subtracting the predicted logarithmic term, the normalized residual approaches a finite constant for each mode. No logarithmic coefficient is fitted. This supports conjecture 6.3, but does not close the gap between the fixed-eigenfunction Rayleigh calculation and an actual-eigenvalue theorem.

7 Low-temperature thermodynamics from the same endpoint data

The finite-temperature TBA and the low-energy gapless interpretation of the negative-spin model are established in Ref. [17]. We use that framework, with the Yang–Yang thermodynamic convention [25], to derive the next temperature coefficient and connect it to the scalar endpoint functions. This section sets $\kappa = 1$, so the bare energy is

$$e_0(x) = -\frac{2}{1+x^2}. \tag{7.1}$$

The free energy and the dimensionless positive function $E = 2\pi R - 2$ should not be confused: the physical ground-state energy at this coupling is $-E$.

7.1 Dressed energy, sound velocity, and compressibility

The zero-temperature dressed energy satisfies

$$(I - K_Q)\epsilon_0 = e_0 - \mu, \qquad \epsilon_0(x) = -2\pi\rho_Q(x) - \mu f_Q(x), \qquad \epsilon_0(Q) = 0. \tag{7.2}$$

Therefore

$$\mu = -\frac{2\pi\eta}{\chi} = -\frac{\pi D'}{\chi^2}. \tag{7.3}$$

Differentiating along the family of ground states gives $\partial_Q\epsilon_0(x) = -\mu' f_Q(x)$, because the boundary terms are proportional to $\epsilon_0(\pm Q) = 0$. With $u = \epsilon_0'(Q) > 0$, where this prime is the rapidity derivative in the definition of u, one obtains $u = \mu'\chi$. Thus

$$v_s = \frac{u}{2\pi\eta} = -\frac{\mathrm{d}}{\mathrm{d}Q}\log(-\mu), \qquad K_{\mathrm{Lutt}} = \chi^2, \qquad \mathcal{K} := \frac{\partial D}{\partial\mu} = \frac{\chi^2}{\pi v_s}. \tag{7.4}$$

All formal weak-coupling sectors of these quantities follow by ordinary series differentiation and division. No new endpoint seed is required.

For example,

$$\mu_0(v) = -4\pi v\left[1 + 2v + \left(\frac{9}{2} - \frac{\pi^2}{6}\right)v^2 + O(v^3)\right], \tag{7.5}$$

$$v_{s,0}(v) = 2\pi v\left[1 + 3v + \left(8 - \frac{\pi^2}{3}\right)v^2 + O(v^3)\right]. \tag{7.6}$$

These formulas describe the Fermi-point dispersion, not the full physical excitation spectrum at arbitrary momentum.

7.2 A quartic Sommerfeld coefficient

The nonlinear thermal equation and grand potential are

$$\epsilon_T(x) = e_0(x) - \mu - T\int_{\mathbb{R}} K(x-y)\log(1 + e^{-\epsilon_T(y)/T})\mathrm{d}y, \tag{7.7}$$

$$\Omega(T,\mu) = -T\int_{\mathbb{R}} K(x)\log(1 + e^{-\epsilon_T(x)/T})\mathrm{d}x. \tag{7.8}$$

Write

$$\Omega(T,\mu) = \Omega_0(\mu) + \omega_2(\mu)T^2 + \omega_4(\mu)T^4 + O(T^6), \qquad \omega_2 = -\frac{\pi}{6v_s}. \tag{7.9}$$

The quadratic coefficient is the standard one-component gapless result. Its complete formal weak-coupling series is inherited from equation (7.4).

For the next term, put $c_1 = \pi^2/6$, $c_2 = 7\pi^4/360$, and $\epsilon_T = \epsilon_0 + T^2\epsilon_2 + \cdots$. Sommerfeld expansion at the two Fermi points yields

$$(I - K_Q)\epsilon_2(x) = -\frac{c_1}{u}[K(x-Q) + K(x+Q)], \tag{7.10}$$

so

$$\epsilon_2(x) = -\frac{c_1}{u}[\mathscr{R}_Q(x,Q) + \mathscr{R}_Q(x,-Q)]. \tag{7.11}$$

Use the local energy coordinate $z = \epsilon_0(x)$ near $x = Q$, with $\partial_z = (\partial_x \epsilon_0)^{-1}\partial_x$. The result is

$$\boxed{\omega_4 = -2\left[\frac{\rho\epsilon_2^2}{2\partial_x\epsilon_0} - c_1\partial_z\left(\frac{\rho\epsilon_2}{\partial_x\epsilon_0}\right) + c_2\partial_z^2\left(\frac{\rho}{\partial_x\epsilon_0}\right)\right]_{x=Q}}. \tag{7.12}$$

The density and dressed energy in this expression are continued smoothly through the endpoint by their integral equations. Self-adjointness cancels the otherwise unknown fourth-order dressed-energy correction. The derivation and the reduction of all endpoint derivatives to $D, \chi, \mathcal{G}$ are given in appendix E. Thus equation (7.12) is both a numerical endpoint formula and a coefficient-generating rule for its formal weak-coupling trans-series.

At $Q = 0.9$, the independent ground-state calculation gives

$$\mu = -1.2999150573992906, \qquad v_s = 0.6138856180302584, \qquad \omega_4 = -0.6292090698625181. \tag{7.13}$$

Solving equation (7.7) independently, define

$$C_4(T) = \frac{\Omega(T,\mu) - \Omega_0(\mu) + \pi T^2/(6v_s)}{T^4}. \tag{7.14}$$

Table 8 shows convergence to the endpoint prediction. The tests use ordinary floating-point thermal solves with resolution checks; unlike table 3, they are not interval certificates.

Table 8: Independent thermal validation at $Q = 0.9$. The canonical calculation adjusts μ to keep the particle density fixed. The two quartic limits are different.

T	Grand-canonical $C_4(T)$	Canonical quartic estimator
0.02000	—	−0.60106621374
0.01000	−0.62971106824	−0.59948400796
0.00500	−0.62933435272	−0.59909158066
0.00250	−0.62924040225	—
0.00125	−0.62921705415	—
Predicted $T \to 0$	−0.62920906986	−0.59896161401

7.3 Fixed-density conversion

Because $D = -\partial_\mu\Omega$, a fixed-density expansion requires

$$\mu(T) = \mu_0 + \frac{\partial_\mu\omega_2}{\mathcal{K}}T^2 + O(T^4). \tag{7.15}$$

Performing the Legendre transform gives

$$F(T,D) = F_0(D) + \omega_2 T^2 + \left[\omega_4 + \frac{(\partial_\mu\omega_2)^2}{2\mathcal{K}}\right]T^4 + O(T^6). \tag{7.16}$$

At the point in equation (7.13), the canonical quartic coefficient is −0.5989616140053465, and the quadratic chemical-potential shift is $-0.2153506235547108\,T^2$. The thermal density residuals are near machine precision. In both ensembles the discrepancy of the quartic estimator decreases approximately as T^2, as expected from the next T^6 term in the free energy.

7.4 High-density behavior and a nontrivial cancellation

Expanding equation (7.12) in the running coupling gives

$$\omega_4(v) = -\frac{1}{2880v^3} + O(v^{-2}), \qquad \omega_4(Q) \sim -\frac{\pi^3}{360}Q^3. \tag{7.17}$$

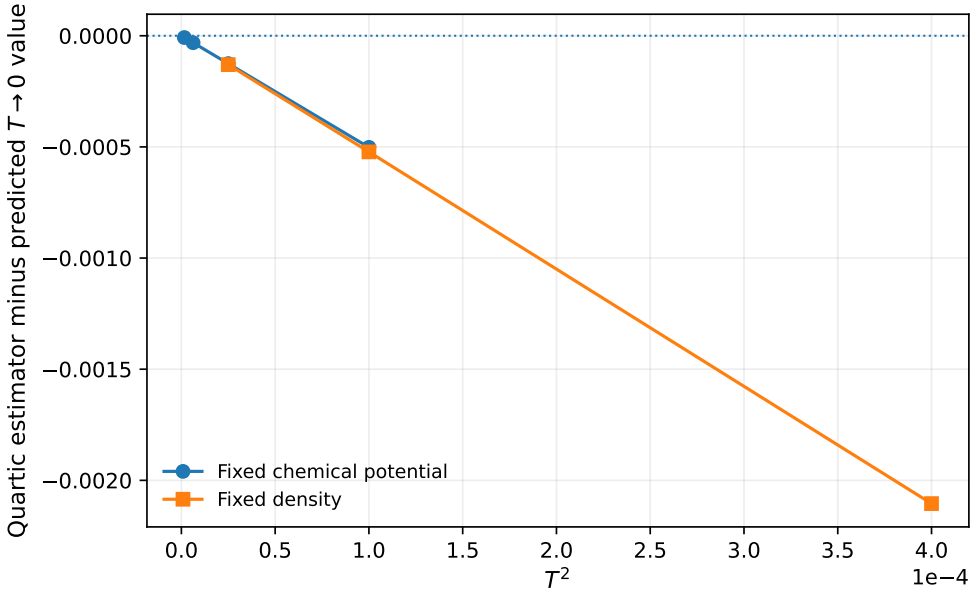


Figure 10: Grand-canonical and fixed-density validation of the quartic temperature coefficient at $Q = 0.9$. Each plotted quantity has its own predicted $T \to 0$ value subtracted. The approach is approximately linear in T^2, which tests the expected next-order remainder.

The potentially larger v^{-4} terms cancel. The three terms in equation (7.12), including the overall factor, contribute respectively

	v^{-4}	v^{-3}
1	$-1/82944$	$1/9216$
2	$1/82944$	$-7/27648$
3	0	$-7/34560$

(7.18)

and sum to equation (7.17). This cancellation is checked symbolically, rather than inferred by fitting a few large-Q data points.

The weak-coupling sectors of ω_2 and ω_4 are formally accessible, but the thermal tests here verify their full finite-Q values, not individually exponentially small thermal sectors. Moreover, a low-temperature expansion at fixed Q is not automatically uniform when Q tends to infinity and the sound velocity vanishes.

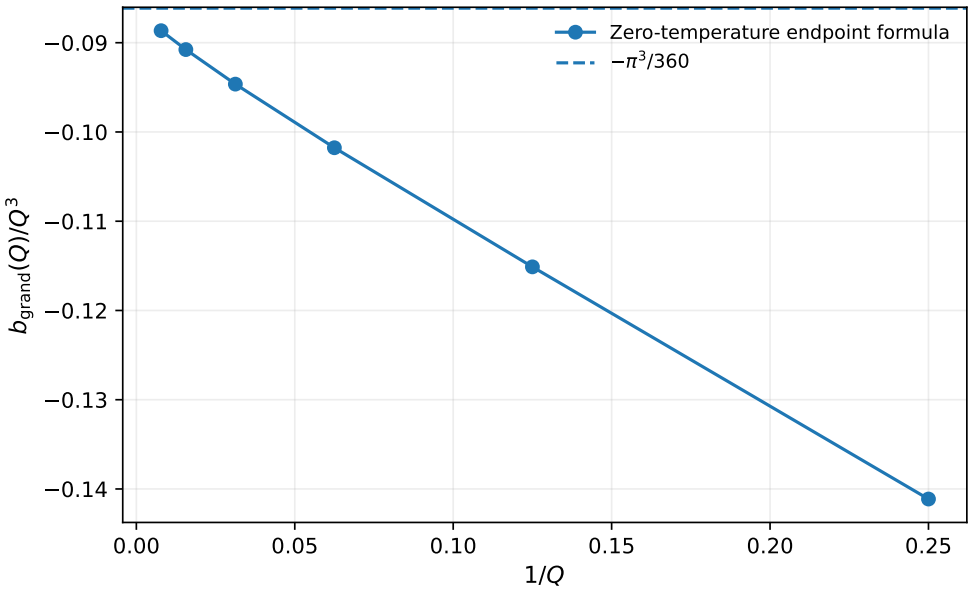


Figure 11: High-density quartic coefficient from the zero-temperature endpoint formula. The limiting line is the analytic value $-\pi^3/360$. The calculation tests the cancellation of the naively larger Q^4 contribution. It does not establish a uniform simultaneous low-T, high-Q limit.

8 Discussion and well-posed remaining problems

The continuation goes beyond a candidate exponential scale in three different ways. First, it provides an explicit formal endpoint construction and an all-sector matrix formula, with high-order coefficient generation and independently resolved subleading sectors. Second, exact interval-variation identities determine the determinant and the full lattice profiles from the boundary function. Third, the deformed determinant has a controlled fractional-spectral scaling limit that is stronger than a single leading eigenvalue asymptotic.

These results do not all have the same logical status. The boundary reconstruction and regularized determinant limit are finite-interval or scaling theorems with proofs given above. The endpoint spectral multiplier is a formal construction, not a proved global Borel reconstruction. The first logarithmic actual-eigenvalue correction is a prediction supported by a derived Rayleigh logarithm and four-mode data. The trans-series comparisons are extensive numerical evidence, with a certified reference but an empirically assessed resummation error. These distinctions are part of the result rather than qualifications to be removed by extrapolation.

8.1 One boundary function and the propagation of Stokes data

The exact Goursat representation changes the main analytic question. The point observables have no independent exact functional freedom once χ is known. Can equation (2.15) be used to derive the endpoint multiplier equation (3.14), the initial response equation (3.16), and the external residues equation (3.33) directly from the fixed-source boundary trans-series?

A proof would need uniform continuation and growth bounds that permit asymptotic and Borel operations under the characteristic double integral. The estimate equation (2.19) proves convergence on finite rectangles but does not justify extending a large-Q expansion to the origin. This is the principal missing step between the proposed endpoint construction and a global reconstruction theorem. It would also explain structurally why the boundary determinant contains only even powers of Λ, while the central observables admit odd ones.

8.2 Uniform spectral resurgence near a zero

The correct spectral object is now fixed:

$$\mathfrak{D}_Q(u) = \det{}_2(I - uA^{-1}) + \text{finite-}Q\text{ corrections.} \tag{8.1}$$

The meaningful next problem is to compute its algebraic, logarithmic, and exponential deformations uniformly near $u = \lambda_j$, then solve for the moving zeros. Evaluating $F(z, Q)$ only at $z = 1$ loses precisely the information needed

for this step. The exact sum equation (6.13) is a stringent normalization condition for any proposed two-variable construction.

Before resolving an exponentially small eigenvalue sector, one must control the perturbed endpoint eigenfunctions well enough to prove or revise conjecture 6.3. The data do not justify assigning a physical mass gap to these operator eigenvalues. Nor does a Borel singularity identify an attractive Hamiltonian or its Bethe vacuum. An attractive-dual interpretation would require those model choices explicitly; none is supplied here.

8.3 High density and low temperature

The cancellation in equation (7.17) shows that naive endpoint power counting overestimates the high-density quartic coefficient. It is natural to seek a uniform solution of the nonlinear thermal TBA in a simultaneous low-T, high-Q limit, retaining the distinction between fixed density and fixed chemical potential. Two Sommerfeld coefficients do not determine its scaling function or even suffice to prove a common domain of uniformity. This is a genuinely thermodynamic question, not merely a request for more terms of the zero-temperature series.

8.4 A probabilistic meaning of the nonperturbative sectors

The killed-Cauchy-walk interpretation is already central to Ref. [1] and has classical antecedents in the capacitor problem [26, 27]. Let τ be the first exit step from the interval. For z within the resolvent domain, the first-step recurrence gives

$$\mathbb{E}_x[z^\tau] = 1 - (1 - z)(I - zK_Q)^{-1}1(x). \tag{8.2}$$

Its derivative at $z = 1$ gives the mean exit time, $f_Q(x)$; in particular, $D + 1$ is the central mean exit time. The zeros of $F(z, Q)$ control singularities of the generating resolvent, subject to possible cancellations for a particular source or evaluation.

This supplies a well-defined alternative to an unspecified physical duality: can the finite-step corrections to the Cauchy-process limit, including exponentially small terms, be associated with identifiable overshoot or repeated-boundary processes? Such an interpretation would connect the density, the spectral determinant, and the origin of distinct observable-dependent sector rules.

8.5 Conclusion

For the three principal lattice observables, the remaining computational obstruction is no longer an unspecified endpoint function. There is a finite formal algorithm, high-order data, and nontrivial numerical validation. The determinant is even more economical because it needs only the constant-source boundary function. At the exact level, one boundary function reconstructs the full problem; at the spectral level, one regularized determinant controls all fixed low modes.

The strongest unresolved task is analytic rather than numerical: proving that the endpoint formal construction is the Borel representation of the exact boundary-to-bulk solution. Resolving another tiny sector would improve evidence, but would not by itself supply that proof.

Data and reproducibility. The companion archive contains the LaTeX manuscript, vector figures, plotting code, all displayed data, three computational modules with their retained coefficient files and parameter scans, and fresh audit logs. Representative endpoint and determinant resummations, interval reference calculations, and all three fast test suites were rerun for this manuscript. The expensive 210-order flow-generation record is retained rather than silently described as a new benchmark. Appendix G gives the exact organization and commands.

Attribution of supplied material. The O(3) commuting-flow representation is taken from the unpublished note by Dániel Nógrádi [14]. The endpoint extension and the present synthesis are distinguished from that input. The unpublished note itself is not included in the public supplementary archive.

A The formal endpoint spectral calculation

This appendix makes the formal step behind equation (3.16) explicit. At zero odd-zeta times a generalized Jost solution compatible with equation (3.9) is

$$a_k^* = \sqrt{t}\left(\frac{1+t}{2}\right)^k e^{ke^{-2u}/2}\,{}_2F_0\left(\frac{1+k}{2}, \frac{1+k}{2}; ; \frac{w}{k}\right), \qquad t = \tanh u = \sqrt{1-2w}. \tag{A.1}$$

Here ${}_2F_0$ is a formal asymptotic series. Direct substitution verifies $(a_k^*)'' + 2k(a_k^*)' = U_* a_k^*$ coefficient by coefficient. The zero-source amplitude is treated separately; naive substitution of $k = 0$ is not a uniform limiting procedure.

Writing the hypergeometric factor through Tricomi's U converts the source inversion into a spectral contour. On its central cut the phase is

$$\Phi(y) = \frac{y+1/2}{w} + \frac{1}{2}\log\left|\frac{y}{1+y}\right|. \tag{A.2}$$

Its perturbative root is $y = -1/2$, with $\Phi'(-1/2) = (1-2w)/w$. The central Jacobian and amplitude give $h_{\mathrm{cl}} = 2\sqrt{w}/t^{3/2}$. The other roots approach the cut ends exponentially and are not included in the power-series seed. Dressing the central contribution with the even spectral multiplier gives $h_* = \mathcal{E}(L_*)h_{\mathrm{cl}}$.

The two multiplier calculations used here are elementary:

$$m(k)m(-k) = \pi\frac{\Gamma(1+k/2)\Gamma(1-k/2)}{\Gamma((1+k)/2)\Gamma((1-k)/2)} = \frac{\pi k}{2}\cot\frac{\pi k}{2}, \tag{A.3}$$

$$[\log m(k)]_{\mathrm{odd}} = \sum_{j\geq 1}\frac{4^j-1}{(2j+1)4^j}\zeta(2j+1)k^{2j+1} = -\sum_{j\geq 1}4^j\tau_{2j+1}k^{2j+1}. \tag{A.4}$$

The first odd term is $\zeta(3)k^3/4$, a useful sign check. The odd part agrees with the spectral phase of the ordered commuting-flow operators.

This calculation identifies a formal central contribution. It does not provide a uniformly controlled contour deformation separating that contribution from every endpoint term, or the Borel growth estimates needed as the source label tends to zero. The exact reconstruction in theorem 2.2 offers a route to such a proof. Neither algebraic consistency nor high-precision agreement is presented as a substitute for it.

B Coefficient algorithm and sector arithmetic

A calculation has two cutoffs: a perturbative order N and an exponential level $n_{\max}$. The finite algorithm is as follows.

1. Generate $U_*, a_{0,*}, h_*$ and $w(v)$ to guarded order. Retain the flow structure rather than expanding all independent odd-zeta monomials.
2. Integrate the commuting flows to equation (3.11); extract A_{00}, D_0, R_0 with equation (3.20). Fix the known leading constants and check the gauge against the independent n_e, n_r input.
3. Generate a_r, X_r, I_r and the symmetric N_{rs} needed by the exponential budget, using equations (3.27) to (3.30).
4. Expand $M = P + PNP + \cdots$ only through $\Lambda^{n_{\max}}$. Assemble equations (3.35) and (3.36), and independently recover R with equation (3.43).
5. For the determinant apply equation (5.3); for thermodynamics apply the scalar endpoint identities, converting Q- and b-derivatives consistently.

Scalar series use half-integer Laurent exponents. A dictionary key k means $v^{k/2}$, with complex coefficients stored as pairs of decimal strings. The explicit R_0 logarithm is stored separately. Negative powers are not passed into a Borel transform. Tiny guarded-order numerical remnants are not interpreted as additional leading physical terms.

The first-order inversion equation (3.31) costs linear time in N after the right-hand side is formed. Products and second-order amplitude recursions incur convolution cost. At fixed $n_{\max}$ only finitely many labels and paths occur. We do not claim an asymptotically optimal implementation of the scalar flow stage.

For an independent algebraic audit put $\kappa = \mathrm{diag}(2, 4, 6, \ldots)$. Then

$$\dot{P} = -\kappa P - P\kappa, \quad \dot{c} = -\kappa c, \quad \dot{I} = \kappa I + ah, \quad \dot{X} = \kappa X + a_0 a. \tag{B.1}$$

Differentiating $M^{-1} = P^{-1} - N$ gives

$$\dot{M} = -\kappa M - M\kappa + Maa^T M. \tag{B.2}$$

Consequently,

$$\dot{J} = \kappa J + a(h + a^T c), \qquad \dot{Z} = -\kappa Z + Ma(h + a^T Z). \tag{B.3}$$

These identities prove $\pi\dot{D} = \chi H$ directly. Differentiating the symmetric quadratic form in equation (3.36) gives $2\pi\dot{R} = H^2$. This is a complete algebraic compatibility check, independent of the unresolved analytic source-contour argument.

C Fisher–Hartwig normalization calculation

With $g(t) = (2\pi)^{-1} \int e^{-ipt} \log\sigma(p)\mathrm{d}p$, expand

$$\log(1 - e^{-|p|}) = -\sum_{n\geq 1} \frac{e^{-n|p|}}{n}. \tag{C.1}$$

The cosine integral gives

$$g(t) = -\frac{1}{\pi} \sum_{n\geq 1} \frac{1}{n^2 + t^2} = \frac{1}{2\pi t^2} - \frac{\coth(\pi t)}{2t}. \tag{C.2}$$

Because $tg(t)^2 \sim 1/(4t)$ at infinity, the quadratic integral requires a logarithmic subtraction. After $u = \pi t$, use

$$\int_0^\infty u^{z-1}(\coth u - u^{-1})^2 \mathrm{d}u = 2^{2-z}\Gamma(z-1)(z-3)\zeta(z-1), \quad -2 < \Re z < 0. \tag{C.3}$$

The strip follows from the quadratic behavior at zero and constant limit at infinity. The finite part of the right-hand side at $z = 0$ is

$$\gamma_E - \frac{2}{3} + \log 2 + 12\zeta'(-1). \tag{C.4}$$

Undoing the factor $1/4$ and the coordinate change gives C_E in the main text.

In the single-root Wiener–Hopf/Fisher–Hartwig normalization, with interval length $2Q$,

$$\begin{aligned} \beta_{\mathrm{FH}} &= C_E - \frac{\gamma_E}{4} + 2\log G_B(3/2) + \frac{1}{4}\log 2 \\ &= -\frac{1}{6} + \frac{1}{4}\log(4\pi) + 3\zeta'(-1) + 2\log G_B(3/2). \end{aligned} \tag{C.5}$$

Here G_B is Barnes' function. Its duplication and special-value identities reduce this to equation (5.9). This fixes all constants used in the numerical comparison without fitting.

This calculation is an evaluation within the standard normalization, not a new proof of the general Fisher–Hartwig theorem. A self-contained theorem for this particular symbol should check the limiting regularity and factorization hypotheses after separating its root. The determinant comparison tests the resulting normalization very sensitively but does not replace those hypotheses.

D Spectral trace and logarithmic Rayleigh correction

D.1 Evaluation of the trace integral

The Green-kernel calculation first gives

$$\mathrm{Tr}\, A^{-2} = \frac{8}{\pi^2} \int_0^\infty (u\coth u - 1)\,\mathrm{csch}^2 u [\log\coth(u/2)]^2 \mathrm{d}u. \tag{D.1}$$

After $z = \tanh(u/2)$ let J be the integral in equation (6.15) without $4/\pi^2$. Expanding its nonsingular combination gives

$$J = 2\left[1 + \sum_{n\geq 1} \frac{1}{(2n+1)(2n-1)^3} - \sum_{n\geq 0} \frac{1}{(2n+1)(2n+3)^3}\right] = \frac{8}{3} - 8 \sum_{\substack{m\geq 3 \\ m \text{ odd}}} \frac{1}{m^3(m^2-4)}. \tag{D.2}$$

Use

$$\frac{1}{m^3(m^2-4)} = \frac{1}{32}\left(\frac{1}{m-2} + \frac{1}{m+2} - \frac{2}{m}\right) - \frac{1}{4m^3}. \tag{D.3}$$

The sum equals $13/48 - 7\zeta(3)/32$, so $J = 1/2 + 7\zeta(3)/4$. This proves the exact trace formula.

D.2 Boundary Fourier tails

For normalized ϕ_j with square-root endpoint traces, the leading one-sided Fourier amplitude is proportional to $\Gamma(3/2)|p|^{-3/2}$. The nonoscillatory squared tail is

$$|\widehat{\phi}_j(p)|^2_{\text{nonosc}} \sim \frac{\pi(c_+^2 + c_-^2)}{4|p|^3}. \tag{D.4}$$

For $1 \ll |p| \ll h^{-1}$,

$$\frac{1 - e^{-h|p|}}{h} - |p| = -\frac{h}{2}p^2 + O(h^2|p|^3). \tag{D.5}$$

The resulting window contribution is

$$-\frac{c_+^2 + c_-^2}{8} h \int^{1/h} \frac{\mathrm{d}p}{p} = -\frac{\lambda_j}{2\pi} h \log(1/h) + O(h). \tag{D.6}$$

The complementary frequencies and oscillatory cross-endpoint term contribute at nonlogarithmic order. This establishes the fixed-function Rayleigh coefficient. Uniform control of the actual perturbed endpoint traces is still needed for conjecture 6.3.

Table 9: First-mode residual after the predicted logarithm is subtracted. $C_1(Q) = Q[\lambda_1 - Q(1-\mu_0)] - \lambda_1 \log Q/(2\pi)$. Two mesh spacings are Richardson-extrapolated; the logarithmic coefficient is not fitted.

Q	$Q(1-\mu_0)$	$C_1(Q)$
128	1.142592670854	1.049133769
256	1.149649144283	1.058148747
512	1.153451774391	1.063412354
1024	1.155485137580	1.066445346
2048	1.15656630144	1.068174725
4096	1.157138671506	1.069152321

E Sommerfeld derivation and scalar endpoint closure

The distributional Sommerfeld expansion is

$$T\log(1 + e^{-\epsilon/T}) = -\epsilon \mathbf{1}_{\epsilon<0} + c_1 T^2 \delta(\epsilon) + c_2 T^4 \delta''(\epsilon) + \cdots. \tag{E.1}$$

For $\epsilon = \epsilon_0 + T^2\epsilon_2 + T^4\epsilon_4 + \cdots$, the quartic boundary distribution is

$$\mathcal{N}_4 = \tfrac{1}{2}\epsilon_2^2 \delta(\epsilon_0) + c_1\epsilon_2 \delta'(\epsilon_0) + c_2 \delta''(\epsilon_0). \tag{E.2}$$

The bulk term is $-\epsilon_4 \mathbf{1}_{[-Q,Q]}$, and $(I - K_Q)\epsilon_4 = -K\mathcal{N}_4$. Using $(I - K_Q)\rho = K$ and self-adjointness eliminates ϵ_4 from the grand potential:

$$\omega_4 = -\int_{\mathbb{R}} \rho(x)\mathcal{N}_4(x)\mathrm{d}x. \tag{E.3}$$

The two endpoints contribute equally. Changing to $z = \epsilon_0(x)$ gives equation (7.12), including the minus sign associated with the first derivative of the delta function.

All required rapidity derivatives reduce to scalar endpoint-length derivatives. Put

$$r = -\tfrac{1}{2}\mathcal{G}', \qquad s = \frac{\chi'}{2\chi}, \qquad A_e = r + s, \tag{E.4}$$

where primes in the rest of this appendix mean Q-derivatives. If A_x denotes the first-argument derivative of $\mathscr{R}_Q(x,Q)+\mathscr{R}_Q(x,-Q)$ at $x=Q$, then

$$A_x=\frac{s'}{2}+\frac{s^2-r^2}{2}-rs. \tag{E.5}$$

For either $g=f_Q$ or $g=\rho_Q$, let $\gamma(Q)=g_Q(Q)$. Then

$$g_x=\gamma'-\gamma A_e,\qquad g_{xx}=\gamma''-\gamma' A_e-\gamma(A_e'+A_x). \tag{E.6}$$

Use $\gamma=\chi$ for f and $\gamma=\eta=D'/(2\chi)$ for ρ. The dressed-energy jets are

$$u=\epsilon_{0,x}(Q)=\mu'\chi,\quad \epsilon_{0,xx}(Q)=u'+\mu' f_x,\quad \epsilon_{0,xxx}(Q)=\epsilon_{0,xx}'(Q)+\mu' f_{xx}. \tag{E.7}$$

Also,

$$\epsilon_2(Q)=-\frac{c_1A_e}{u},\qquad \epsilon_{2,x}(Q)=-\frac{c_1A_x}{u}. \tag{E.8}$$

These formulas close the quartic coefficient on $D,\chi,\mathcal{G}$ without rapidity-grid differentiation.

For the canonical conversion write $\mu(T)=\mu_0+\mu_2T^2+\cdots$. The fixed-density constraint gives $\mathcal{K}\mu_2=\partial_\mu\omega_2$. In $F=\Omega+\mu D$ the linear variation cancels, leaving the quartic shift

$$(\partial_\mu\omega_2)\mu_2-\tfrac{1}{2}\mathcal{K}\mu_2^2=\frac{(\partial_\mu\omega_2)^2}{2\mathcal{K}}. \tag{E.9}$$

A fixed-μ thermal solve must therefore not be compared with the canonical quartic coefficient.

F Interval certification of the reference equation

Let y_j,w_j be exact Gauss–Legendre nodes and weights on $[-Q,Q]$, enclosed by interval arithmetic. Computed coefficients c_j are treated as fixed decimal numbers, not as an exact linear-system solution. The analytic trial profile is

$$f_N(x)=g(x)+\sum_j w_jK(x-y_j)c_j, \tag{F.1}$$

where $g=1$ or K. Every node residual $f_N(y_j)-c_j$ is interval-evaluated.

For $0<\delta<1$, the physical ellipse of semiminor axis δ has parameter

$$\varrho_\delta=\frac{\delta}{Q}+\sqrt{1+(\delta/Q)^2},\qquad k_\delta=\frac{1}{\pi(1-\delta^2)}. \tag{F.2}$$

For real $x\in[-Q,Q]$, its analytic integrand obeys

$$|K(x-y)f_N(y)|\le M_\delta=k_\delta\left(g_\delta+k_\delta\sum_j w_j|c_j|\right), \tag{F.3}$$

with $g_\delta=1$ or k_δ. Gauss quadrature is exact through degree $2N-1$. Bounding the remaining Chebyshev coefficients gives

$$\left|\int_{-Q}^{Q}K(x-y)f_N(y)\mathrm{d}y-\sum_j w_jK(x-y_j)f_N(y_j)\right|\le\frac{8QM_\delta\varrho_\delta^{-2N}}{1-\varrho_\delta^{-1}}. \tag{F.4}$$

Add $(2Q/\pi)\max_j|f_N(y_j)-c_j|$ for the collocation residual, then use equation (4.2) for the solution enclosure.

The implementation brackets each Legendre root by a disjoint certified sign change and encloses its weight. At $Q=0.9$, $N=64$, $\delta=0.9$, raw bounds are below 2.44×10^{-47} for D and 1.19×10^{-47} for R. Conservative output radii are 10^{-46}. At $Q=2$, $N=96$, the radii are 10^{-32} and 10^{-33}. Computations use mpmath interval arithmetic at at least 110 decimal digits; the code is not separately formalized in a proof assistant.

This certificate is independent of the trans-series. It certifies neither a finite-order Padé continuation nor a Borel-integral tail nor equality of an infinite sector sum with the exact solution.

G Reproducibility and the evidence ledger

The archive retains three computational modules with their original internal layouts, preserving the relative paths to coefficient files and quadrature caches.

Directory	Contents
`reproducibility/endpoint/`	Flow generator, seeds, point sectors, primitive blocks, Chebyshev TBA, lateral/Green summation and failed-method scans.
`reproducibility/determinant_thermal/`	Determinant sectors, parity Nyström reference, gap solver and thermal checks.
`reproducibility/spectral_reconstruction/`	Goursat reconstruction, spectral determinant, fractional matrix, interval references and canonical thermal test.
`audit/`	Fresh test logs, principal endpoint and determinant rechecks, and interval output.
`data/`, `figures/`, `code/`	Manuscript data extracts, vector plots and figure/table regeneration script.

The source hashes identify the September PDF and the supplied inputs. The unpublished O(3) note by Dániel Nógrádi is retained privately and is not included in the public supplementary archive. Archived notes preserve the calculation's chronology; the present manuscript supersedes their comparisons with the March version.

G.1 Checks and principal resummations

Install `requirements.txt`. Run each module's `run_checks.py` from its own `code` directory:

```
cd reproducibility/endpoint/code
python run_checks.py
cd ../../determinant_thermal/code
python run_checks.py
cd ../../spectral_reconstruction/code
python run_checks.py
```

The principal endpoint test uses the supplied coefficients:

```
# From reproducibility/endpoint/code
python green_resum.py ../data/sectors200_n12.json \
  --Q .9 -T 190 --phi .50 --inner 384 --outer 32 \
  --dps 200 --evaldps 90 --out ../results/green_new.json
python factorized_resum.py ../data/basis200_r8.json \
  --Q .9 --T0 198 --TI 190 --TF 130 --phi .4 \
  --nodes 512 --dps 200 --green ../results/green_new.json \
  --out ../results/factorized_new.json
```

The Green file is specific to Q, and a mismatched value is rejected. The determinant and interval tests are

```
# From reproducibility/determinant_thermal/code
python determinant_numeric.py resum --Q .9 --T 154 \
  --phi .45 --nodes 512 --dps 170
# From reproducibility/spectral_reconstruction/code
python interval_tba.py --Q 0.9 --n 64
```

The optional 210-order seed regeneration uses C++17 and Boost; the commands and numerical precision are specified in the endpoint module's README. Its output is included, so regeneration is not required for the comparisons.

G.2 Fresh audits and retained production records

For this manuscript all three test suites, the principal factorized endpoint sum and its Green replacement, the 154-term determinant sum, and the $Q = 0.9$, $N = 64$ interval certificate were rerun. Figure and table data were regenerated from the records. The full high-order flow and every large spectral or thermal scan were not rerun wholesale; their raw outputs and code are retained, with representative checks in the fresh suites.

The archive provides all complex sector contributions, cumulative sums, angle/order scans, eight limiting fractional modes, four finite-Q gap modes, deformed determinants at several spectral parameters, both thermal ensembles at two densities, and failed conformal tests. Main-text tables select decisive comparisons but expose the unresolved next order. No formal sector beyond the tested levels is described as numerically resolved.

Table 10: Final evidence ledger. Generation is not the same as numerical resolution.

Object	Generated / established	Independent evidence and limitation
D, R, E	Through $n = 12$, powers through v^{200}	D, R resolved through $n = 7$; E follows exactly; eighth point sectors unstable.
$\log F$	Even sectors through $n = 8$, 180 orders	Eighth sector resolved in three scans; no endpoint seed needed.
Boundary reconstruction	Convergent finite-interval theorem	Independent grid test; asymptotic/Borel interchange unproved.
Deformed determinant	Locally uniform fractional limit	Matrix data and exact trace; exponential zero shifts unknown.
Logarithmic gap term	Fixed-function Rayleigh coefficient	Four-mode support; actual-eigenvalue theorem still needed.
Thermal T^4	Both ensembles, endpoint closure	Nonlinear thermal tests; individual thermal NP sectors not isolated.
Reference TBA	Uniform residual enclosures	Certifies reference only, not Borel–Padé error.

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